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Guarding the Gates of Innovation

An Empirical Analysis of Formal Appropriation Mechanisms as Moderators of the Relationship between R&D external sourcing and Radical Innovation Performance

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Abstract

This thesis asks whether formal appropriation strengthens or weakens the innovative return to externally sourced R&D, and whether the answer differs between buying research under contract and co-developing it with a partner. Using the Panel de Innovación Tecnológica, 69,034 firm-year observations on 10,946 Spanish manufacturing firms between 2009 and 2016, the study separates external sourcing by counterparty and appropriation by legal form, and estimates the occurrence of market-novel innovation and the turnover share appropriated from it as distinct outcomes. Both governance modes raise the probability of a market-novel introduction, and that advantage declines as the firm's patent position strengthens; in the market channel it is entirely diminished for firms among the most heavily protected firms. The attenuation does not appear for trademarks and copyright, nor for intra-group or institutional purchase, and it operates on the occurrence of radical innovation rather than on the revenue appropriated from it. The same instrument nonetheless performs its expected function over the firm's own output: within firms, an increase in patent-based protection raises the market-novel share of turnover and lowers the share new only to the firm. Appropriation is therefore neither a general enabler of open innovation nor a general obstacle to it; it secures what a firm generates and diminishes what it draws from others, and the complement to external sourcing is internal research capability, not legal protection.

Keywords: appropriation; patents; R&D outsourcing; technological collaboration; radical innovation; absorptive capacity; PITEC; correlated random effects.

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1. Introduction

In recent decades, firms have increasingly turned to external sources for innovation, rather than relying solely on internal research and development. Chesbrough (2003) documented the shift from closed, vertically integrated research to an open model in which firms routinely draw on external sources, and the pattern has since been established well beyond the cases he studied: across European manufacturing, a substantial share of research and development is bought from suppliers or co-developed with partners rather than performed in-house (Laursen and Salter 2006; Cassiman and Veugelers 2006). By doing so, firms seeking to remain competitive in a globalized economy can access expertise and innovative ideas that would be costly to develop internally, and reach the market with products genuinely new, assembled from recombining knowledge they did not necessarily produce themselves (Tojeiro-Rivero, Moreno and Badillo 2019). However, this strategy carries inherent risks not associated with in-house research. To acquire or share knowledge, firms must disclose certain details about their projects, potentially compromising the very assets they aim to protect (Arrow 1962; Teece 1986).

The literature provides a framework for conceptualizing the problem, yet it does not offer a definitive solution. Transaction-cost economics and property-rights theory suggest that a firm can only safely share its valuable knowledge externally when there are safeguards beyond the contractual agreement that stops a partner from misusing it (Williamson 1985; Grossman and Hart 1986). Formal protection does precisely that by converting disclosed knowledge into a contractible, enforceable asset, so that a firm able to defend a claim over what it discloses can afford to disclose more. On this reading, stronger protection should make external sourcing pay off more. A similar logic extends to collaboration: because a patent makes disclosure safe, it can let a firm work openly with a partner without fear of being expropriated (Arora, Athreye and Huang 2016). The opposite view is the paradox of openness (Laursen and Salter 2014): a guarded, protective stance is the opposite of the open, sharing one that external search depends on, so building up protection should reduce the returns to sourcing rather than raise them. As a third view, the resource-based and relational traditions hold that the knowledge which matters most is tacit and socially complex, precisely what a legal instrument cannot govern. The resource-based view further adds that the significance of external information is contingent upon the firm's internal research capabilities, particularly

its absorptive capacity. This capacity is essential for the firm to effectively interpret and utilize the information it acquires (Cohen and Levinthal 1990).

The initial two arguments present opposing perspectives on the same relationship, each supported by substantial evidence. Both, however, treat appropriation as an explanation of firm decisions: whether to source externally, how widely to search, which governance mode to adopt. Neither estimates whether the firms that source under strong protection obtain more from having done so. The theoretical claim concerns the productivity of external exchange; the empirical work concerns the propensity to engage in it. So while theory holds that outsourcing knowledge close to the firm's core should require the support of intellectual-property rights, there is little systematic evidence on whether such rights in fact raise or lower the innovative return to external sourcing. The link between R&D outsourcing and radical innovation under different regimes of formal protection remains under-explored, and it is under-explored on the margin where firms experience it.

This thesis explores whether firms can effectively mitigate these risks and whether protection instruments like patents truly enable them to benefit from open innovation. The study asks whether a high degree of formal protection strengthens or weakens the association between external R&D sourcing and radical innovation, and whether the answer differs between the two channels through which firms source. That qualification matters: buying research under contract and co-developing with a partner expose a firm to the disclosure hazard differently, and the rival accounts imply different patterns across channels as well as different signs, so examining both permits a discrimination neither channel alone allows. Radical innovation is taken throughout to mean market novelty: introducing, and earning revenue from products that are new to the market, not merely new to the firm. A second question follows: if protection does not raise the return to knowledge acquired externally, does it still perform its expected function over the innovation the firm generates?

The setting is Spanish manufacturing between 2009 and 2016, drawn from the Panel de Innovación Tecnológica (PITEC), the firm-level panel implementing the Community Innovation Survey for Spain. It suits the question because it records the counterparty of each external R&D transaction and the individual instruments a firm uses to protect its innovations, allowing the sourcing construct to be separated by channel and the appropriation construct by legal form.

The thesis presents two main contributions, each with specific boundaries and implications. The first contribution reorients the discussion of appropriation by shifting the focus from whether protection encourages firms to source externally to whether it alters the nature of what they acquire. This shift highlights the divergence between two competing theories, a question not previously addressed directly. The second contribution demonstrates that appropriation is not a singular force; rather, it involves protecting a firm's output and managing its external relationships, which exert opposing influences through the same mechanism. This suggests that the commonly used composite appropriability indices may obscure actual effects, presenting both a measurement and theoretical challenge. By simultaneously estimating governance channels and outcome margins, and employing theory-based placebo tests, the study quantifies the differences between channels and subjects its interpretations to potential falsification. The overall aim is to identify the specific impact of appropriation, demonstrating that it is neither a universal facilitator nor a barrier to open innovation, rather than asserting unsupported causal effects.

The thesis proceeds as follows. Chapter 2 develops the theoretical framework and states each hypothesis with the pattern that would disconfirm it. Chapter 3 describes the data, the variables and the empirical strategy. Chapter 4 reports the estimates and the robustness analysis. Chapter 5 discusses the findings in relation to prior work, sets out the contributions and their policy implications, and states the limitations and directions for further research.

2. Theoretical Framework and Hypotheses

This chapter builds the framework in five steps: first, it establishes the micro-foundation of the disclosure hazard. Second, it isolates two distinct roles that appropriation plays: one over the output of innovation, on which the literature agrees, and one inside the governance of external exchange, on which it divides into two rival accounts with opposite predictions. The third section specifies the governance construct and the fourth partitions the appropriation axis, placing the analytical cut inside formal protection then crosses the two axes into a governance-appropriation alignment matrix. The fifth states the hypotheses, each with the pattern that would disconfirm it, together with the falsification tests that discriminate the proposed mechanism from an artefact.

2.1 Micro-foundations: the information paradox and the disclosure hazard

The information paradox first articulated by Arrow (1962) arises from an asymmetry inherent to the exchange of knowledge as an economic good. For a firm to acquire external knowledge (whether through arm's-length contracting or through a cooperative arrangement) it must first specify what knowledge it needs, and specifying that need requires disclosing enough of its own technological position for the counterparty to determine whether it can supply what is required. The prospective supplier cannot commit to deliver, and cannot price the deliverable, without understanding the buyer's current capability, the gap the external knowledge is meant to fill, and the technical context in which the purchased knowledge will be integrated. This structural feature produces a bidirectional disclosure requirement: the buyer discloses in order to enable the transaction, and the disclosure itself transfers a portion of the very asset the transaction is meant to protect. Teece (1986) reframes the problem as the foundational challenge of appropriation, arguing that whether an innovator profits from its innovation depends jointly on the appropriability regime and on the control of the complementary assets required for commercialisation. Laursen and Salter (2014), building on Cohen and Levinthal (1990), identified the downstream regularity that structures this chapter: the paradox of openness, a concave (inverted U) relationship between a firm's appropriability strategy and the scope of its external engagement, estimated fractionally (Papke and Wooldridge 1996) on 2,931 UK innovators from CIS 2005; the estimation there pools manufacturing and services; this study restricts to manufacturing, for the reason given in Section 3.1. This mechanism is not confined to formal contracting. It operates in cooperative arrangements as well, though there the disclosure is smoothed over repeated interaction rather than concentrated at contract signing. This difference will carry substantial weight in the alignment argument below.

One feature of the hazard must be specified precisely, because it determines the outcome against which the hypotheses are assessed. Leakage does not, in the first instance, undermine the revenue from a product already commercialised under the firm's name. It operates earlier: a supplier or partner who misappropriates the disclosed knowledge may dilute the output, allow a competitor to match the firm's timeline, or enter the product market itself. Each of these bears first on whether a market-novel product is launched under the focal firm's name, and only afterward on the turnover it earns. The disclosure hazard is therefore, in the first instance, a threat to the occurrence of market entry rather than to the return conditional on entry, which is why the framework distinguishes the occurrence of market novelty from the appropriated return to it (Section 2.3).

The intensity of the disclosure requirement varies with the type of knowledge sourced and its position relative to the firm's core competencies. Peripheral knowledge (inputs, standardised components, well-understood processes) can be specified in codified terms without revealing the firm's differentiated position, and can be sourced through the market with limited exposure. Knowledge central to competitive advantage, or expressed as the tacit understanding Polanyi (1966) described, cannot be specified without exposing the underlying capability, and its external sourcing carries a leakage risk proportional to the strategic centrality of the domain in which it is integrated. The resource- and knowledge-based tradition draws the strategic inference from this, treating such knowledge as the basis of durable differentiation precisely because it resists codification and therefore transfer (Barney 1991). This gradient underpins the captive/non-captive distinction of Tojeiro-Rivero and Moreno (2025): intra-group sourcing contains the hazard within structures the firm controls, whereas third-party sourcing exposes it to a hazard that must be managed through the appropriation regime itself. Their finding that non-captive sourcing pays for breakthrough innovation only where the intellectual-property environment is strong is the closest existing gesture toward the mechanism this thesis examines; and it remains a gesture, since the regime enters their argument as context, never as a firm-level condition on the return.

2.2 Two roles for appropriation, and two rival accounts of the second role

The theories of the firm that bear on the problem converge on one role for appropriation, which is around the decisions to protect the output and divide on the other role regarding appropriation inside the governance relationship, and it is the division that supplies the analytical engine of the chapter.

2.2.1 The settled role: appropriation over the output

Property-rights theory (Grossman and Hart 1986; Hart and Moore 1990; Hart 1995) and the Schumpeterian separation of invention from commercialisation (Schumpeter 1934; Teece 1986) agree that an enforceable claim over a completed, market-novel technology enables a firm to defend its margin against imitators and shifts the composition of its revenue toward the appropriable segment. This is a claim about the composition of output, not about the governance of exchange. Its principal complication is informative: Hurmelinna-Laukkanen, Sainio and Jauhiainen (2008) find a strong appropriability regime raising the incremental share of turnover while showing no significant effect for radical innovations. That null sits on the very margin this thesis measures, and the framework reads it as a composition effect

awaiting decomposition rather than as evidence that appropriation is inert at the radical margin. From the enforcement side, Fan et al. (2024) find stronger intellectual-property enforcement raising productivity, concentrated among larger, already-competitive firms.

2.2.2 The contested role: appropriation inside the governance relationship

The disagreement concerns whether protection raises or lowers the productivity of external exchange itself, and two accounts predict opposite signs on the same relationship.

The alignment account. Transaction-cost economics and property-rights theory (Coase 1937; Williamson 1975, 1985; Grossman and Hart 1986) hold that a patent mitigates the hazard of opportunism in arm's-length outsourcing by converting the commissioned knowledge into an enforceable, contractible asset, raising the returns to outsourcing: the deliverable becomes legally as well as technically defined, recourse exists if the supplier defects, and the buyer can disclose enough for the supplier to work because what is disclosed is already fenced. On the relational side, the **secure-openness** argument holds that patents can be facilitators of openness rather than obstacles to it. By providing the legal security that makes disclosure safe, a patent lets a firm share tacit knowledge with a partner without fear of expropriation, and firms can deliberately use protection to design the terms on which they open up (Arora and Ceccagnoli 2006). The empirical footprint is real: Arora, Athreye and Huang (2016) find open technology leaders patenting far more than closed ones, with no analogous gap among followers, and Spithoven and Teirlinck (2015) and Veugelers and Cassiman (1999) find that a strong protection posture accompanies, rather than deters, engagement in external sourcing.

The substitution account. The paradox of openness and its relational-friction mechanism (Laursen and Salter 2014; Miozzo, Desyllas, Lee and Miles 2016; Hurmelinna-Laukkanen and Puumalainen 2007) hold that a protective posture contradicts the open, sharing posture that effective external search and co-development require. A patent-heavy stance signals an intent to appropriate, raises the coordination cost of every joint decision about ownership, and impedes the openness on which the absorption of external knowledge depends. The resource and knowledge-based and relational views (Barney 1991; Teece, Pisano and Shuen 1997; Dyer and Singh 1998) supply the reason: the tacit, socially complex knowledge that co-development transfers is precisely what a legal instrument cannot govern, since what cannot be codified cannot be claimed (Langlois, BenMahmoud-Jouini and Servajean-Hilst 2023).

Table 2.1. The literature divergence: two accounts of appropriation inside the governance relationship

	Alignment account	Substitution account
Theoretical source	Transaction-cost economics and property rights (Williamson 1985; Grossman and Hart 1986); secure openness (Arora and Ceccagnoli 2006)	Paradox of openness and relational friction (Laursen and Salter 2014; Miozzo et al. 2016); resource-based and relational views (Barney 1991; Dyer and Singh 1998)
Mechanism	Protection converts disclosed knowledge into a contractible, enforceable asset and shifts residual control to the buyer	Protection signals an intent to appropriate, raises the coordination cost of joint decisions, and deters the partners that make openness valuable
Predicted sign, market row	Positive: contractibility is most valuable where recourse rests on contract alone	Negative: a protective posture is incompatible with the openness external search requires
Predicted sign, relational row	Positive, through a different mechanism: patents appropriate jointly created output and permit disclosure without expropriation	Negative, through the same mechanism: the account is not channel-specific and therefore predicts a common attenuation
Cross-channel pattern implied	Positive in both rows, magnitudes free to differ because the mechanisms differ	Negative in both rows, magnitudes expected to be similar because the mechanism is common

Own elaboration from the cited literature.

The two accounts are not opposites to be chosen between wholesale; the paradox of openness holds them together. Laursen and Salter (2014) show the relationship between protection and productive collaboration to be concave: moderate protection facilitates openness in the manner the secure-openness view describes, while the relational-friction penalty binds at high legal aggressiveness rather than for every patenting firm, so the sign of the patent-collaboration relationship is not fixed a priori but depends on where firms sit on that curve. What the reviewed literature does not do is estimate appropriation as a condition on the governance-to-performance relationship : in the alignment literature, appropriation sits on the right-hand side of an equation whose outcome is a governance choice; in the substitution literature, appropriation sits on the right-hand side of an equation whose outcome is the breadth of openness. The divergence is genuine, unadjudicated, and unresolved on precisely the margin where firms experience it.

2.3 Two governance modes of external knowledge sourcing

External R&D sourcing is not a single behaviour (Tojeiro-Rivero and Moreno 2019, 2025). R&D outsourcing is a market transaction: the firm specifies a deliverable, agrees a price, and receives the output under contract, so it favors codified knowledge that can be described in advance. Technological collaboration is a relational arrangement in which partners co-develop and iterate, the channel through which tacit knowledge travels. The distinction is behavioural, not merely descriptive: Tojeiro-Rivero and Moreno (2019) find that the returns to cooperation rise with a region's tacit-knowledge endowment while the returns to outsourcing fall with it: two modes that respond to the same condition with opposite signs are not two intensities of one behaviour. For the Spanish setting, Miguélez, Moreno and Artís

(2011) further support the interpretation of internal research intensity as the capability connecting a firm to its knowledge environment.

2.3.1 Three outsourcing channels, only one of which carries the hazard

Counterparty type is fundamental of the outsourcing construct, not a matter of robustness, because the leakage hazard is live only where the counterparty both competes in the buyer's product market and lies outside the buyer's ownership boundary. Non-captive commercial sourcing (arm's-length purchase from firms outside the corporate group) is the channel in which the disclosure hazard operates without institutional containment, and it is the channel Tojeiro-Rivero and Moreno (2025) find raises the market-novel sales share. Captive sourcing from within the corporate group settles residual control by ownership rather than contract, so the hazard that appropriation instruments exist to bound is already bounded; this channel serves as a falsification check. Institutional sourcing from universities and public research organisations involves counterparties governed by publication norms and standard transfer contracts who do not compete downstream, a population for which the argument makes no prediction. Tojeiro-Rivero, Moreno and Badillo (2019) confirm the relevance of counterparty identity on the same data: R&D acquired from abroad raises radical but not incremental innovation, the effect weaker for university than business-sector knowledge. Their result concerns cross-border acquisition, whereas the construct here pools domestic and foreign arm's-length purchase. The pooling is deliberate: the hazard follows from the counterparty lying outside the ownership boundary and competing downstream, which a domestic competitor satisfies as fully as a foreign one. Cross-border enforcement asymmetry may still make it more severe abroad, a moderator this design does not estimate.

2.3.2 The two economic margins of the outcome

Because the disclosure hazard threatens market entry before it touches revenue, the outcome is understood along two economic margins that the hypotheses treat separately: the occurrence of a market-novel launch, and the appropriated share of turnover it earns once launched. The two margins answer to different forces. Occurrence is a developmental question: any real external sourcing extends the firm's recombinant reach and should raise the probability of a market-novel launch, subject to a hollowing-out limit at which heavy externalisation weakens the internal capability that absorption requires (Kotabe 1989; Grimpe and Kaiser 2010). The appropriated share is a profitability question: sourcing raises it only past a genuine commitment threshold, below which a marginal purchase has neither

committed enough to reshape the portfolio nor disclosed enough to incur real leakage, so heavy sourcing pays where light sourcing does not (Tojeiro-Rivero and Moreno 2025; Mol 2005). The translation of this developmental/profitability distinction into functional form is further discussed to Chapter 3.

2.3.3 Functional form: the construct is dose-dependent

Tojeiro-Rivero and Moreno (2025) find that non-captive sourcing raises the market-novel share only above a substantial (above-median) commitment threshold. Below the threshold there is no effect to moderate, and a binary indicator that treats one per cent and sixty per cent identically averages a real effect over a population in which most treated observations are inert. Mol (2005), on 52 Dutch industry observations, makes the same point from a different direction in finding that the stock and the flow of outsourcing tell opposite stories about its relationship with internal R&D intensity.

This framework does not predict where the threshold falls; its location is an empirical question the pre-estimation must answer, and the chapter commits only to the prediction that a threshold exists and that the effect below it is null.

2.4 The appropriation axis: patents versus the rest, flow versus stock

The appropriation construct is partitioned as patent-based versus non-patent, a cut placed inside formal protection, for three reasons. The legal reason is that patents and utility models confer an enforceable, disclosure-based claim over specific technical content which is exactly the claim relevant to a commissioned technology and to the returns of a market-novel product (Teece 1986; Grossman and Hart 1986; Arora and Ceccagnoli 2006). On the other hand, trademarks and copyright protect identity and expression, not technical content, and so should carry no appropriation story in this setting. The empirical reason is not that patents are the most effective mechanism. It has been proven in the literature that lead time and secrecy usually outrank them (Levin et al. 1987; Cohen, Nelson and Walsh 2000; Arundel 2001); but that patents are the instruments whose legal form fits the questions this study asks. The instrumental reason is that the informal family varies far less across firms and functions as a near-universal baseline rather than a discriminating strategy (Thomä and Bizer 2013; Miric, Boudreau and Jeppesen 2019), so the margin on which firms genuinely differ, and on which observation is possible, is the patent margin.

This further splits into the stock and the flow of patent protection. These are strategic constructs, not statistical artefacts. The patent stock is the firm's accumulated, path-dependent body of protected knowledge, in other words it is the cumulative record of its appropriation strategy, built up over time and changing only slowly, in the tradition that treats knowledge as an accumulating capital stock (Griliches 1979) and innovation as path-dependent (Dosi 1982; Nelson and Winter 1982). A large stock is a firm-type marker whose association with radical performance the defensive-patenting literature reads as ambiguous or negative, because habitual, portfolio-scale patenting is often blocking and freedom-to-operate activity rather than the protection of fresh novelty (Hall and Ziedonis 2001; Cohen, Nelson and Walsh 2000). The two constructs are identified from different sources of variation, and the empirical strategy that separates the accumulated position from the current move is set out in Chapter 3 defined along with the remaining list of constructs and measures.

2.5 Deriving the alignment matrix from the divergence

The matrix is not a taxonomic device adopted for expositional convenience. It is derived from the divergence set out in 2.2 crossing the two axes into a matrix that integrates the moderating hypotheses.

Expressing appropriation as a condition on the governance-to-performance relationship, on which the accounts disagree, is (to the knowledge of the author) a novel contribution of this study. This study proposes that arbitration between the two accounts requires two channels rather than one, because the accounts differ not only on sign but on cross-channel pattern: alignment predicts a positive moderation in both channels with magnitudes free to differ, since it reaches them through different mechanisms (contractibility in the market channel, secure openness in the relational one), while substitution predicts a negative moderation of similar magnitude in both, its mechanism being channel-independent. A two-channel design also admits a third pattern that neither account predicts on its own and that a single interaction would render invisible: a positive market-row moderation coexisting with a negative relational-row moderation, which would indicate that contractibility and relational friction each operate, on their own rows, and that appropriation is neither a general complement nor a general substitute for external exchange.

Crossing the two modes with patent use against a non-patent reference yields the four cells of Table 2.2, the identifying contrasts running along the rows: Cell A against Cell B for the market channel, Cell C against Cell D for the relational one. The framework does not

reconcile the accounts in advance; it commits to reading a negative moderation as a confirmation of the substitution account, and states the meaning of each outcome before it is tested.

Table 2.2. Governance mode by patent use: the alignment and substitution predictions

	Patent-based protection	Non-patent protection (reference)
R&D outsourcing (non-captive, market governance)	Cell A. Alignment predicts a patent premium: patents create the contractible ex ante boundaries that arm's-length exchange requires, and residual control shifts to the buyer. Substitution predicts a penalty, through the paradox of openness.	Cell B. Reference for H1: an informed but unbound counterparty, with recourse resting on contract alone. Trademarks and copyright confer no claim over the commissioned technology (Arrow 1962).
Technological collaboration (relational governance)	Cell C. Alignment, in its secure-openness form, predicts a premium: collaborative exposure is continuous and directed at a frequently horizontal partner, and patents appropriate the jointly created output. Substitution, through relational friction, predicts a penalty.	Cell D. Reference for H2: tacit co-development under non-patent protection, where the protective repertoire is an evolving practice rather than an ex ante codification (Polanyi 1966; Dyer and Singh 1998; Langlois et al. 2023).

Own elaboration. Each cell carries two opposite-signed predictions; the design arbitrates between them.

Non-patent formal instruments confer no claim over technical content and should therefore moderate neither channel; the test that matters is not each interaction against zero but patents against non-patent instruments, and a failure to distinguish them would refute the appropriation reading rather than support a weaker version of it (F1). Captive and institutional sourcing already contain or neutralise the leakage hazard and should carry no patent moderation on the margin where the effect is real (F2). The theoretical warrant for F2 is the same evidence that motivates the channel decomposition: the asymmetry reported by Tojeiro-Rivero, Moreno and Badillo (2019).

2.6 Absorptive capacity as the complement

Where appropriation and external sourcing may act as substitutes, one factor is theorised as their complement: absorptive capacity, the internal research capability through which external knowledge is recognised, assimilated and applied (Cohen and Levinthal 1989, 1990; Zahra and George 2002). The substitution reading that must be overcome is the make-or-buy reading, in which sourcing and protection are alternatives: a firm confident in its appropriation regime dares to source externally, while a firm that cannot protect stays closed, which is close to how Spithoven and Teirlinck (2015) interpret their finding that protection predicts outsourcing intensity, and how the tradition running from Veugelers and Cassiman (1999) through Mol (2005) frames the choice. That reading rests on a hidden assumption: that purchased knowledge deploys itself. It does not. Cohen and Levinthal (1990) established that recognising, assimilating and applying external knowledge requires a capability built through

the firm's own research, and Zahra and George (2002) decomposed that capability into potential and realised dimensions, the second of which (transformation and exploitation) is where sourced knowledge becomes sold product. Once absorption enters, sourcing, appropriation and capability are sequential stages of one value chain and are complements in the sense of Milgrom and Roberts (1990, 1995), each raising the marginal return to the others; substitution survives only below the capability threshold at which the downstream stage fails. Absorptive capacity is treated as a between-firm characteristic rather than as a short-run choice variable the firm can adjust (Escribano, Fosfuri and Tribó 2009); Arbussa and Coenders (2007) add, for the Spanish setting, that it is two-dimensional, scanning capability helping all firms while integration capability helps only those that protect.

2.7 Hypotheses

The hypotheses are stated in the language of economic incentives and property rights, each with the pattern that would disconfirm it. A Schumpeterian distinction disciplines the shared outcome: invention and commercialisation are distinct processes with distinct outputs (Schumpeter 1934), and novelty is graded by reference to the market rather than to the firm, following the convention the data implement (OECD and Eurostat 2005). The new-to-market share of turnover is accordingly read as the appropriated return to radical innovation and the share from products new only to the firm as the incremental comparison, as in Hurmelinna-Laukkanen, Sainio and Jauhiainen (2008) and, on the same data, Tojeiro-Rivero, Moreno and Badillo (2019). Radical innovation throughout means market novelty, not firm-level adoption.

The baseline: the two margins of external sourcing

Both governance modes are expected to raise the occurrence of market-novel innovation, subject to a hollowing-out limit at high externalisation; on the profitability margin, external sourcing raises the appropriated share only above a commitment threshold. The baseline is an associational claim, its identifying variation lying largely in differences across firms rather than changes within them.

Baseline. *R&D outsourcing and technological collaboration are each positively associated with the occurrence of market-novel radical innovation (extensive margin); the appropriated share of turnover rises only above a commitment threshold in sourcing intensity (intensive margin).*

H1 and H2: appropriation on the governance relationship

The pre-registered prediction is the alignment one: patent protection raises the return to each governance mode, and the framework specifies in advance that a negative moderation is not a null but a confirmation of the substitution account: a protective posture crowding out the openness external search requires. Because the substitution mechanism concerns the incompatibility of postures rather than the structure of disclosure, it predicts a common attenuation across both channels. In the relational channel the prediction is explicitly two-sided: the secure-openness view expects patents to facilitate collaboration by making disclosure safe, while relational friction expects a penalty at high legal aggressiveness, so H2 tests the net sign along the concavity of the paradox of openness rather than assuming it.

H1. *Patent-based protection positively moderates the effect of R&D outsourcing on radical innovation (alignment); the pre-specified alternative is a negative moderation (substitution, through the paradox of openness).*

H2. *Patent-based protection positively moderates the effect of technological collaboration on radical innovation (secure openness); the pre-specified alternative is a negative moderation at high legal aggressiveness (relational friction), the net sign depending on the intensity of the protective posture.*

H3a and H3b: appropriation on the output

The output role divides along the stock/flow partition. A current act of patenting attaches to a current innovation and should raise its appropriated share which is supported by the property-rights and Schumpeterian claim (H3a). The disconfirming pattern for H3a is a null or negative within-firm coefficient once firm effects are present.

The defensive-patenting literature reads a habitually high stock as ambiguously or negatively associated with radical performance; the second prediction is stated with a weaker evidential claim, because a negative association with a share outcome is consistent with defensive patenting but also with almost any habitual innovation activity, so a confirming result is not by itself evidence of the defensive-patenting mechanism, and the chapter commits in advance to saying so. The disconfirming pattern is a positive between-firm coefficient.

H3a. *Within firms, a current increase in patent-based protection is positively associated with the appropriated share of turnover from market-novel products.*

H3b. *Between firms, a habitually high stock of patent-based protection is negatively associated with the appropriated share.*

H4: margin specificity

If the mechanism is what the framework claims (a disclosure hazard that threatens novelty rather than production, and an instrument whose legal form fits technical content) the appropriation effects should concentrate on the market-novel margin and attenuate on the share of turnover from products new only to the firm, which is less tacit and less disclosure-sensitive. Tojeiro-Rivero, Moreno and Badillo (2019) provide the direct precedent, finding externally acquired R&D raising radical but not incremental innovation on the same data. A pattern of equal magnitude on both margins disconfirms the radical-specific reading and would indicate a general revenue-composition effect.

H4. *The appropriation effects of H1, H2 and H3a concentrate on the new-to-market margin and attenuate on the new-to-firm margin.*

H5: absorptive capacity as complement

Where internal research capability is high, external knowledge is transformed into market-novel revenue; where it is low, external acquisition substitutes for a capability the firm lacks and the downstream stage fails. Absorptive capacity should therefore amplify the return to external sourcing, on the very margin where the framework anticipates patents may attenuate it: the moderation that Laursen and Salter (2014) and Spithoven and Teirlinck (2015) assert in prose and never estimate. A flat or declining effect of sourcing across levels of internal research capability disconfirms it.

H5. *Absorptive capacity positively moderates the effect of external R&D sourcing on the appropriated share.*

2.8 The conceptual model

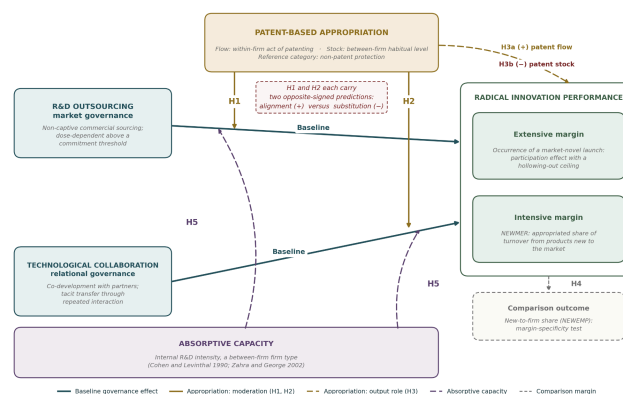


Figure 2.1. The governance-appropriation framework: two sourcing modes, patent-based appropriation in its moderating and output roles, and absorptive capacity as complement.

Three features of the figure 2.1 carry analytical weights and should be read correctly. The first is that the moderating arrows are drawn onto the governance paths rather than into the outcome, because the contested claim concerns the productivity of external exchange and not the level of performance. The second is that only the H1 and H2 arrows are labelled with two signs, since these alone are the contested moderations: the framework is committed in advance to reading a negative coefficient there as confirmation of the substitution account rather than as a failure to find an effect, whereas the output roles and the complementarity prediction each carry a single directional expectation. The third is that the outcome is drawn as two boxes rather than one: the disclosure hazard theorised in Section 2.1 threatens the occurrence of market entry in the first instance and the appropriated share only subsequently, so a framework that collapses the margins would be testing a proposition it has not made. Table A2.4 summarises the hypotheses and their disconfirming patterns

2.9 Scope conditions and maintained assumptions

The framework is bounded by three scope conditions: it makes no claim about average effects: nothing in the argument contradicts the finding that legal and strategic instruments show similar main effects, because the argument lives in the interactions (Arbussà and Coenders 2007); it does not require that firms pair protection and governance deliberately, the effect operates whether the combination was chosen or inherited; and the patent/non-patent partition is a first-order approximation, defended as the sharpest cleavage the data can support rather than the only one available (Miozzo et al. 2016; Thomä and Bizer 2013). Three assumptions are maintained rather than tested: that the pairing of governance and appropriation is a visible manifestation of an underlying core-versus-non-core allocation firm-level data cannot resolve directly; that the Spanish intellectual-property regime is stable enough across the window to treat firm-level posture as the operative source of variation (Fan et al. 2024); and that appropriation postures move slowly relative to annual sourcing decisions. Claims identified within firms carry greater weight than those identified across them: the governance baseline is stated associationally, the output claims more strongly, and because identification rests on the panel rather than an exogenous shock, every estimate is a conditional association, not a causally identified effect. The estimation is set out in Chapter 3.

3. Data, Measures and Empirical Strategy

3.1 Data and sample

The empirical material is the Panel de Innovación Tecnológica (PITEC), the firm-level innovation panel assembled by the Spanish National Statistics Institute with FECYT and the COTEC Foundation. PITEC implements the Community Innovation Survey instrument on a rotating panel of Spanish firms under the definitions of the Oslo Manual, so its core constructs are comparable with CIS-based evidence from other European economies.

The analysis covers manufacturing firms from 2009 to 2016 (unbalanced panel) and was executed with STATA18. The scope restriction to manufacturing follows the argument of Chapter 2: the disclosure hazard the framework describes operates on technological knowledge embodied in products, and patents and utility models are a meaningful protective option in manufacturing in a way they are not across services. The window begins in 2009 because every regressor is measured at $t-1$, which sacrifices the first observed wave (2003), and because the counterparty-composition items on which the channel separation depends are not consistently available in the earliest waves (2004-2008). The dataset was cleaned, missing observations dropped and sector harmonised. The window closes in 2016 as per data availability. All time-varying regressors are lagged one period; the lag is built once, as a set of pre-lagged variables shared by every model, so that all specifications read the same lagged quantity. It is not offered as an identification strategy: it removes simultaneity within the reference period but does not address time-varying confounding, and no causal claim rests on it. The two-margin design implies the estimation samples that are not interchangeable (see Table A3.1 in the appendix)

The extensive-margin sample is every manufacturing firm-year with complete data. The intensive-margin sample conditions on a market-novel launch having occurred, since the new-to-market share is undefined for firms with no such product, and the comparison sample is identical in composition but carries the new-to-firm share instead. Two smaller sub-samples are used only where the relevant variable is required, and estimates from the latter are not generalised beyond the IP-active population. These samples include firms with data on the qualifications of their research staff, and firms holding at least one formal instrument.

3.2 Variable operationalisation

Constructs are defined below by the role they play; variable names follow the PITEC nomenclature, and Appendix Table A3.2 restates the definitions in tabular form.

Dependent variables. *Novedad* indicates whether the firm introduced a product new to the market. *Newmer* is the percentage of turnover such products generate, bounded at zero and one hundred, and is read as the appropriated return to radical innovation. *Newemp*, the turnover share from products new only to the firm, is the comparison margin for H4. The two margins are estimated separately based on the rationale discussed in Section 2.3.2.

Independent variables. Governance enters as two modes, since the two respond to the same contextual variation with opposite signs (Tojeiro-Rivero and Moreno 2019). The treatment (*noncap_hi*) indicates R&D purchase from firms outside the corporate group at above the median intensity among firms that source at all, intensity being such purchase as a percentage of total innovation expenditure. Two measurement decisions are important: The denominator is total innovation expenditure rather than internal R&D, which would confine the sample to R&D performers and would place internal R&D in the denominator of the treatment and in the principal proxy for absorptive capacity at once, building in a mechanical correlation. The median is computed within positive sourcers rather than across all firms, so that the contrast separates heavy from light commitment rather than sourcing from not sourcing, as the dose-dependence argument of Section 2.3.3 requires. (*Coop*) indicates cooperation on innovation with an external partner. (*Captive*) is intra-group purchase and (*inst*) is purchase from universities and public research organisations.

Moderators. *Patent_prot*, a count in [0,2] of patent and utility-model use, carries both roles the framework assigns to appropriation: moderator in H1 and H2, direct output claim in H3. The split inside formal protection because registration is not the mechanism, excludability over technical content is. A trademark confers no claim over a commissioned invention. *Other_formal*, the equivalent count of trademark and copyright use, is therefore a pre-specified null and supplies the F1 placebo. Section 3.4 further separates the within-firm deviation of *patent_prot* from its firm mean, since an accumulated regime is a slowly changing attribute while a discrete act attached to an output is a flow. *Ln_abscap*, log of one plus internal R&D over turnover, moderates in H5 and serves throughout as control and conditioning attribute. Breadth (*formal_breadth*) and composition (*patent_share*) are carried

for robustness only, each correlating with its own components by construction, so neither enters alongside the counts it summarises. *Abscap2_pct*, the graduate share of R&D personnel, is used in a sub-sample check.

Controls. *Ln_size* is entered because larger firms hold the complementary assets commercialisation requires (Teece 1986) while posting mechanically smaller shares from any one product, so a negative sign on the share outcome is expected. *Grupo* captures internal capital and knowledge access, entered separately from captive sourcing to avoid confounding the two. *Export_pp*, intra-EU sales as a percentage of turnover, proxies competitive selection and is expected to be positive. *Pubfund_any* absorbs subsidised activity that would otherwise load onto the sourcing variables. Market-scope indicators (*mdonac*, *mdoue*, *otropais*) control for the breadth of market against which novelty is assessed, a definitional requirement rather than a behavioural control. Two-digit industry and year effects complete the specification.

3.3 Empirical strategy

The outcome is split into two margins, and each is estimated with a method suited to its support. The extensive margin (occurrence) is estimated by probit on the market-novel introduction, and the intensive (new-to-market sales share) and comparison (new-to-firm sales share) margins by random-effects Tobit on the turnover share, censored at zero and one hundred because the outcome is a bounded percentage with mass at both limits. Every model is estimated in correlated random-effects form, entering the firm-specific time means of all time-varying regressors alongside the regressors themselves (Mundlak 1978; Chamberlain 1984).

Model Specification

Let i index firms and t years. All regressors are measured at $t-1$. $w_{i,t-1}$ collects the remaining controls together with year and two-digit industry effects, and $\bar{w}_i$ the firm-specific time means of every time-varying regressor, computed on the estimation sample. The extensive margin is a probit on the occurrence of a market-novel introduction, $d_{it} = 1[d_{it}^* > 0]$:

$$(1) \quad d_{it}^* = \alpha_0 + \alpha_1 \text{noncap_hi} + \alpha_2 \text{coop} + \alpha_3 p + \alpha_4 (\text{noncap_hi} \cdot p) + \alpha_5 (\text{coop} \cdot p) + w' \alpha_6 + \bar{w}'_i \theta^\alpha + \varepsilon_{it}.$$

The intensive margin is estimated by two-limit Tobit. The new-to-market turnover share y_{it} is a bounded percentage with mass at both limits, so a latent-variable model censored at zero

and one hundred is appropriate; the share is defined only where $d_{it} = 1$, which is the selection issue addressed below:

$$(2) \quad y_{it}^* = \beta_0 + \beta_1 \text{noncap_hi} + \beta_2 \text{coop} + \beta_3 p + \beta_4 (\text{noncap_hi} \cdot p) + \beta_5 (\text{coop} \cdot p) + \beta_6 (\text{noncap_hi} \cdot \ln_abscap) + w'\beta_7 + \bar{w}'_i \theta^\beta + c_i + u_{it},$$

$$y_{it} = 0 \text{ if } y_{it}^* \leq 0; \quad y_{it} = y_{it}^* \text{ if } 0 < y_{it}^* < 100; \quad y_{it} = 100 \text{ if } y_{it}^* \geq 100.$$

where p denotes `patent_prot` and subscripts $i, t-1$ are suppressed for readability. Both equations are estimated in correlated random-effects form, which is what the firm-mean terms $\bar{w}'_i \theta$ do. The firm effect is not assumed independent of the regressors but is modelled as a linear function of them, $c_i = \psi + \bar{w}'_i \theta + a_i$, with a_i independent of the regressors by construction (Mundlak 1978). Random effects without these terms would attribute to external sourcing what belongs to the kind of firm that sources. The two θ vectors are estimated separately and are superscripted accordingly. The quantities the hypotheses are read from are the following. Table A3.3 in the appendix presents the coefficient glossary.

Both models are non-linear, so the interaction coefficient is not the interaction effect (Ai and Norton 2003). What is reported throughout is the marginal effect of each governance mode evaluated at each level of protection: on the extensive margin $\partial \Pr(d = 1) / \partial \text{noncap_hi} = \phi(\cdot)(\alpha_1 + \alpha_4 p)$, and correspondingly for collaboration. These are the three points plotted in Figure 4.6 for $p = 0, 1, 2$; the sign of α_4 fixes the slope of that profile, and the pre-specified disconfirming pattern for H1 is a profile that fails to rise.

On the intensive margin the marginal effect is scaled by the probability of being inside the censoring limits, so it is smaller than the latent-index coefficient and is reported for the observed share rather than for y^* .

The new-to-market share is observed only for firms that launched a market-novel product, and that launch is itself the extensive-margin outcome. A two-stage correction is therefore estimated as a diagnostic. The first stage is equation (1) with the exclusion restriction `nodemand` added, p again denoting `patent_prot` and the $i, t-1$ subscripts suppressed:

$$(3) \quad d_{it}^* = \alpha_0 + \alpha_1 \text{noncap_hi} + \alpha_2 \text{coop} + \alpha_3 p + \alpha_4 (\text{noncap_hi} \cdot p) + \alpha_5 (\text{coop} \cdot p) + w'\alpha_6 + \bar{w}'_i \theta \alpha + \gamma \cdot \text{nodemand} + \varepsilon_{it}.$$

The inverse Mills ratio λ_{it} is computed from its fitted index. The second stage is equation (2) with that term added, estimated on the launchers only:

$$(4) \quad y_{it}^* = \beta_0 + \beta_1 \text{noncap_hi} + \beta_2 \text{coop} + \beta_3 p + \beta_4 (\text{noncap_hi} \cdot p) + \beta_5 (\text{coop} \cdot p) + \beta_6 (\text{noncap_hi} \cdot \ln_abscap) + w'\beta_7 + \bar{w}'_i \theta \beta + \rho \cdot \lambda_{it} + c_i + u_{it}.$$

λ_{it} is the expected selection error for a firm that is selected in, so adding it removes that expectation from the outcome equation, and ρ is proportional to the correlation between the two equations' unobservables. A ρ indistinguishable from zero indicates that the uncorrected estimates are unbiased on this account (Heckman 1979; Semykina and Wooldridge 2010).

The restriction matters because λ_{it} is a non-linear function of the first-stage index and can in principle be computed without one. Where both equations carry the same regressors, however, it is close to a linear combination of them and the second stage becomes unstable (Puhani 2000). *Nodemand* indicates that the firm cited insufficient market demand as a reason for not innovating. It records a judgement that no market exists for a novel product, which bears on whether a launch is attempted but, once one has occurred, carries no further information about the share of turnover it earns. Exclusion cannot be verified directly: a firm in a persistently weak market might both refrain from innovating and earn less from what it does introduce.

The specification is also what makes the stock/flow decomposition possible, and this is the substantive reason for adopting it. Fixed effects would remove the same correlation between the firm effect and the regressors, but would discard the between-firm variation H3b is about. Here the coefficient on the level of patent protection is the within-firm effect. This is the flow, and the test of H3a. The coefficient on the firm mean equals the between-firm effect minus the within-firm effect. Taking firm means of both sides of equation (2) shows that the between-firm effect is the sum of the two coefficients; that sum is the stock in H3b. Two quantities follow from this. The firm-mean coefficient tests formally whether stock and flow differ, which is the Hausman condition. A claim about how habitually protecting firms differ from others requires the sum, estimated by linear combination so that the covariance between the two coefficients enters its standard error. H3a is therefore read from the level coefficient and H3b from the sum. The equality of the within-firm coefficient with its fixed-effects counterpart is retained as a diagnostic, reported in the Chapter 4 appendix.

The correlated-random-effects (Mundlak 1978) and two-stage selection design adopted here follows established practice on innovation-survey data: the Mundlak device applied to Spanish and European innovation panels by Tojeiro-Rivero and Moreno (2019) and Tojeiro-Rivero, Moreno and Badillo (2019), whose joint use of both on PITEC is the immediate precedent for the present specification. The Heckman-type selection correction was applied to R&D-sourcing outcomes by Spithoven and Teirlinck (2015) and, in its Semykina-Wooldridge (2010) panel form.

Because an interaction fitted on a misspecified main effect is uninterpretable, the shape of the sourcing-outcome relationship on each margin is established non-parametrically before any moderator is read. Decile and ventile profiles, a scan of the treatment effect across candidate cutoff percentiles, a continuous-dose specification on the identical sample, and smoothed profiles estimated separately for firms that do and do not hold patents together fix the functional form and locate where in the intensity distribution protected and unprotected firms diverge. The cutoff of record (the median among sourcers) is fixed by rule in advance rather than chosen from the scan, whose role is only to show that the finding does not depend on that choice. These profiles are reported in Chapter 4 and its appendix.

Before estimation, the covariance structure of the regressor set is inspected. A bivariate correlation matrix with significance levels is reported at the appendix, together with variance inflation factors for the linear benchmark, so that any collinearity among the governance, appropriation and capability blocks is visible before the interactions are read.

3.4 Hypothesis architecture and placebos

The hypotheses and their disconfirmation criteria are stated in Chapter 2 and are not restated here. The two falsification tests are built into this architecture. F1 replaces patent-based protection with non-patent formal protection, which confers no claim over technical content; since trademark and copyright registrants resemble patent registrants on observables, a moderation of comparable size there would show that the estimate captures a firm type rather than a protection mechanism. F2 replaces the arm's-length treatment with intra-group and institutional sourcing, where the hazard is contained by ownership or by publication norms. Both are assessed by a formal equality test against the corresponding treatment interaction rather than by comparing significance across separate models. Because a null is only informative where the identifying cell is populated and the moderator varies within firms, cell counts accompany every interaction and nulls on thin cells are read as limits on power rather than as findings.

3.5 What the design supports

The design identifies conditional associations, not causal effects. These data offer no instrument, discontinuity or staggered treatment, and the one defensible exclusion restriction is spent at the selection stage, where the requirement it must meet is weaker. What the design does support, it supports carefully: claims read from variation within firms carry more weight

than those read from differences across them, and the estimates are interpreted accordingly. The full set of limitations arising from the data, the measures and the population is set out in Chapter 5, once the estimates that give them their force are in hand.

4. Results

4.1 Descriptive statistics

Table 4.1. Descriptive statistics, both estimation samples

Variable	Extensive mean	SD	Intensive mean	SD
Market-novel product introduction (0/1)	0.253	0.435	1.000	0.000
New-to-market sales share (%)	7.436	20.221	29.355	31.158
New-to-firm sales share (%)	10.455	24.647	13.096	19.510
High-intensity non-captive R&D purchase (0/1)	0.066	0.248	0.109	0.312
Any non-captive R&D purchase (0/1)	0.132	0.338	0.252	0.434
Intra-group R&D purchase (0/1)	0.040	0.195	0.074	0.262
Institutional R&D purchase (0/1)	0.123	0.329	0.239	0.427
Technological collaboration (0/1)	0.282	0.450	0.524	0.499
Patent-based protection (0–2)	0.149	0.421	0.331	0.588
Non-patent formal protection (0–2)	0.158	0.387	0.290	0.496
Breadth of formal protection (0–4)	0.307	0.674	0.621	0.903
Composition: patent share of instrument set	0.449	0.417	0.511	0.403
Absorptive capacity: $\ln(1 + \text{internal R\&D intensity})$	0.044	0.132	0.088	0.178
Log employees	4.208	1.692	4.304	1.611
Business group member (0/1)	0.438	0.496	0.482	0.500
Public innovation funding (0/1)	0.285	0.451	0.525	0.499
Intra-EU sales (% of turnover)	12.992	22.800	16.393	23.863
Firm-years	69,034		17,486	
Firms	10,946		4,796	

All explanatory variables measured at $t-1$. The extensive sample comprises all manufacturing firm-years with complete data; the intensive sample conditions on a market-novel product introduction.

Table 4.1 reports means and standard deviations for both estimation samples. The extensive-margin sample comprises 69,034 firm-years covering 10,946 manufacturing firms; the intensive-margin sample comprises the 17,486 firm-years and 4,796 firms that introduced a market-novel product.

First, only a small proportion of firms receive the treatment: high-intensity non-captive purchase covers 6.6 per cent of extensive-margin firm-years, which bounds the precision attainable for any interaction estimated on it. Second, every governance and appropriation variable approximately doubles in prevalence when the sample is conditioned on a

market-novel introduction: the treatment from 0.066 to 0.109, collaboration from 0.282 to 0.524, patent-based protection from 0.149 to 0.331. This is the descriptive signature of selection on the extensive margin and motivates the diagnostic reported in Section 4.2. Third, the identifying variation is predominantly cross-sectional: 22.0 per cent of firms ever change treatment status and 40.8 per cent ever change collaboration status, so the estimates speak more to differences between firms than to within-firm transitions. Appendix table A4.2 reports the correlation matrix and variance inflation factors. Collinearity among the governance, appropriation and capability blocks is unproblematic; the breadth and composition measures correlate with their own components by construction and are never entered alongside them.

4.2 Estimation, selection, and functional forms

The extensive margin is estimated by probit on market-novel product introduction. The intensive and comparison margins are estimated by random-effects Tobit with limits at zero and one hundred.

The correlated random-effects specification is required formally: a joint test on the firm means rejects the plain random-effects null, so the firm effect is correlated with the regressors and the between-firm and within-firm components must be estimated separately. The specification also reproduces the fixed-effects benchmark to within 0.036 across all coefficients when the firm means are computed on the estimation sample, against a maximum discrepancy of 1.067 when computed on a wider construction window. All models of record therefore use estimation-sample means. Appendix tables Appendix Tables A4.3 to A4.5 report the specification search, the artificial-regression test and the fixed-effects comparison. Figures A4.1, A4.2, A4.3 confirm the threshold choice.

Because the intensive outcome is observed only for firms that introduced a market-novel product, and that introduction is the extensive-margin outcome, a two-stage correction was estimated as a diagnostic. Table A4.7 reports the first stage. The retained exclusion restriction is an indicator for whether the firm cited insufficient market demand as a reason for not innovating. It predicts selection at the one per cent level and does not reject exclusion from the outcome equation ($p = 0.320$). The correction is not carried into the models of record because the inverse Mills ratio is insignificant, ranging from -0.553 to -0.981 against standard errors near 2.5, throughout. The estimates barely move, and the correction costs

precision, as the final two rows of Table A4.7 show. The models below are therefore uncorrected, with the corrected estimates retained as robustness (Table A4.14, R3).

Functional form of each margin

The functional form of each margin was established before any moderator is read, and the theoretical framework predicts a participation effect with a hollowing-out ceiling on the extensive margin and a commitment threshold on the intensive margin.

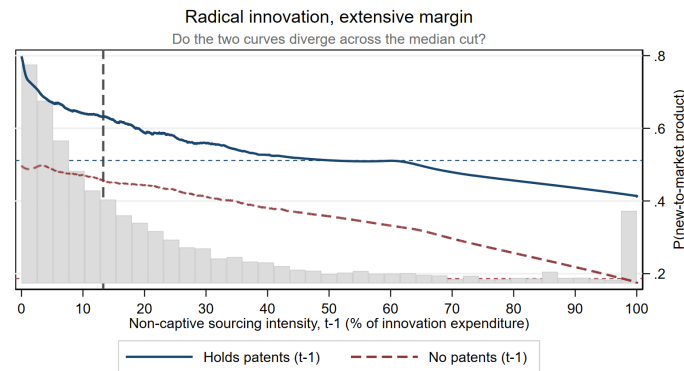


Figure 4.4. Probability of a market-novel launch, by sourcing intensity and patent status.

Figure 4.4 shows that both profiles decline monotonically across the support, from 0.79 to 0.42 among patent holders and from 0.51 to 0.18 among non-holders. The participation premium is each curve's vertical distance from the non-sourcer mean of its own patent status, not the gap between curves; it is larger and more durable where no patent-based instrument is held, in the region of the support where the distribution is dense; that distance is the participation premium the moderation is stated about. Among unprotected firms the premium is 0.33 at the lowest positive intensity, 0.26 at the median cut, and closes only near seventy per cent of innovation expenditure. Among protected firms it is 0.28, falls to 0.14 by the cut, and reaches zero near sixty per cent. One alternative reading should be set aside. In a probit, the effect of sourcing depends on where a group starts, not only on how it responds, so identical behaviour can produce different effects at different baseline probabilities. Here that works against the finding: protected firms start near one half, where the curve is steepest and effects are magnified; unprotected firms start near 0.18, where it is flatter. The attenuation is obtained despite a baseline difference that should have pushed the other way.

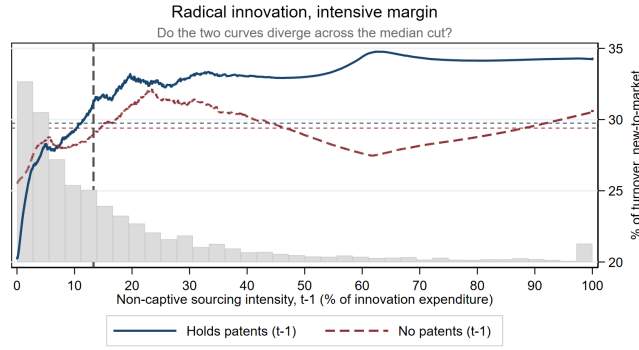


Figure 4.5. New-to-market turnover share, by sourcing intensity and patent status

Figure 4.5 shows that the non-sourcer means are almost equal at 29.5 and 29.0. Both profiles sit below that reference under ten per cent, cross between ten and fourteen, and remain above thereafter. The crossing falls at the median cut, fixed by rule in advance, so the threshold of Section 2.3.3 is recovered where the treatment assumes it. Across the populated region the profiles are indistinguishable, 30.5 against 29.5 at the cut and within a point to forty per cent. The separation above forty per cent runs in the alignment direction but holds few firm-years, is recovered by no specification here, and survives none of the cutoffs; nothing is built on it.

4.3 Main results: appropriation as a governance moderator

Table 4.8. Extensive margin: baseline, H1, H2 and the falsification tests

Dependent variable: market-novel product introduction (0/1)	(1) Baseline	(2) H1, H2	(3) F1 placebo	(4) F2 placebo
High-intensity non-captive R&D purchase	0.093*** (0.028)	0.149*** (0.030)	0.162*** (0.031)	0.149*** (0.030)
Technological collaboration	0.234*** (0.019)	0.263*** (0.020)	0.267*** (0.020)	0.261*** (0.020)
Patent-based protection (0-2)	0.085*** (0.020)	0.179*** (0.024)	0.172*** (0.024)	0.183*** (0.024)
Purchase $\times$ patent-based protection [H1]		-0.208*** (0.039)	-0.188*** (0.041)	-0.206*** (0.039)
Collaboration $\times$ patent-based protection [H2]		-0.150*** (0.025)	-0.141*** (0.027)	-0.140*** (0.027)
Non-patent formal protection (0-2)	0.086*** (0.021)	0.084*** (0.021)	0.104*** (0.025)	0.083*** (0.021)
Purchase $\times$ non-patent protection [F1]			-0.068 (0.046)	
Collaboration $\times$ non-patent protection [F1]			-0.027 (0.030)	
Intra-group R&D purchase	0.121*** (0.042)	0.121*** (0.042)	0.121*** (0.042)	0.121*** (0.044)
Intra-group purchase $\times$ patents [F2]				-0.000 (0.048)
Institutional R&D purchase	-0.003 (0.025)	-0.001 (0.025)	-0.000 (0.025)	0.010 (0.027)

Dependent variable: market-novel product introduction (0/1)	(1) Baseline	(2) H1, H2	(3) F1 placebo	(4) F2 placebo
Institutional purchase $\times$ patents [F2]				-0.035 (0.030)
Firm-years	69,034	69,034	69,034	69,034

Probit estimated in correlated random-effects form with firm means computed on the estimation sample. All regressors at $t-1$. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Absorptive capacity, firm size, business group, public funding, intra-EU sales, market-scope indicators, industry effects and year effects included but not reported.

The baseline is supported on the extensive margin. Both governance modes raise the probability of a market-novel introduction: 0.093 for high-intensity non-captive purchase and 0.234 for collaboration, both significant at one per cent. Appendix table A4.6 reports the non-parametric intensity profiles, which confirm that the effect is non-monotonic in sourcing intensity: the marginal effect declines across the distribution and is non-significant in the highest ventile, consistent with the hollowing-out mechanism of Section 2.3.3.

H1 is disconfirmed in its alignment form and the substitution alternative is supported.

The interaction between high-intensity non-captive purchase and patent-based protection is -0.208 with a standard error of 0.039. Marginal effects are reported in Figure 4.6: the effect of high-intensity purchase on the probability of a market-novel introduction is +0.040 among firms holding no patent-based instrument, -0.018 at one instrument and -0.090 at two. The pre-specified disconfirming pattern for the alignment account was a marginal effect that fails to rise across levels of patent protection. That pattern obtains, and the decline is monotone.

H2 is likewise disconfirmed in its alignment form. The interaction between collaboration and patent-based protection is -0.150 with a standard error of 0.025. The marginal effect of collaborating runs +0.077, +0.049 and +0.010 across the three levels of the moderator. The point estimate remains positive throughout the observed range, though the interval at two instruments contains zero.

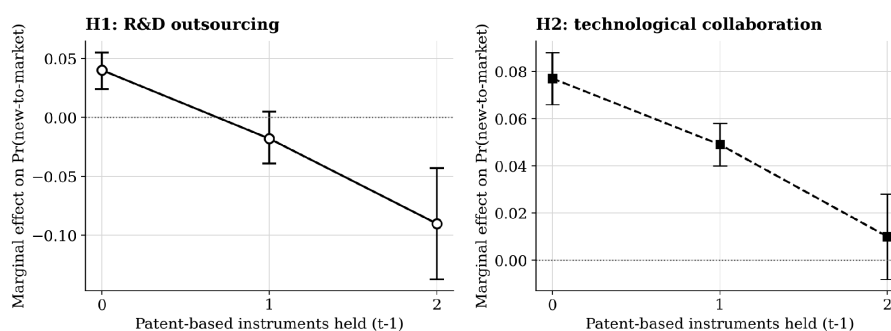


Figure 4.6. Marginal effect of each governance mode on $Pr(\text{new-to-market})$ across levels of patent-based protection, with 95 per cent confidence intervals.

Both falsification tests are passed, one with a qualification. Non-patent formal instruments confer no claim over technical content and should moderate neither channel. Column 3 returns -0.068 (SE 0.046) for the purchase interaction and -0.027 (SE 0.030) for the collaboration interaction, both non-significant, while the patent interactions in the same specification remain -0.188 and -0.141. Column 4 returns -0.000 (SE 0.048) for the intra-group interaction and -0.035 (SE 0.030) for the institutional interaction.

Table 4.9 Intensive margin: baseline, the contested moderations, the falsification and the complementarity tests

Dependent variable: new-to-market sales share (%)	(1) Baseline	(2) H1, H2	(3) F1	(4) F2	(5) H5
High-intensity non-captive R&D purchase	1.823** (0.797)	2.018** (0.914)	2.075** (0.972)	2.014** (0.914)	1.323 (0.979)
Technological collaboration	-0.656 (0.748)	-0.869 (0.816)	-0.668 (0.849)	-0.872 (0.817)	-0.750 (0.858)
Patent-based protection (0-2)	1.606** (0.707)	1.305 (0.909)	1.200 (0.917)	1.419 (0.929)	1.322 (0.910)
Purchase × patent-based protection [H1]		-0.537 (1.123)	-0.498 (1.190)	-0.488 (1.124)	-0.675 (1.125)
Collaboration × patent-based protection [H2]		0.555 (0.816)	0.786 (0.861)	0.669 (0.848)	0.560 (0.816)
Non-patent formal protection (0-2)	0.980 (0.654)	0.950 (0.657)	1.489 (0.908)	0.944 (0.657)	0.942 (0.657)
Purchase × non-patent protection [F1]			-0.174 (1.372)		
Collaboration × non-patent protection [F1]			-0.823 (0.973)		
Intra-group R&D purchase	-1.250 (1.088)	-1.265 (1.088)	-1.259 (1.088)	-0.250 (1.234)	-1.275 (1.088)
Intra-group purchase × patents [F2]				-2.380* (1.361)	
Institutional R&D purchase	-1.002 (0.682)	-1.021 (0.682)	-1.011 (0.682)	-1.053 (0.779)	-1.007 (0.682)
Institutional purchase × patents [F2]				0.136 (0.880)	
Absorptive capacity	14.405*** (2.968)	14.305*** (2.973)	14.430*** (2.977)	14.396*** (2.974)	14.742*** (3.835)
Purchase × absorptive capacity [H5]					8.309** (4.160)
Collaboration × absorptive capacity [H5]					-1.836 (3.390)
Inverse Mills ratio	-0.725 (2.531)	-0.981 (2.573)	-0.764 (2.585)	-0.848 (2.576)	-0.620 (2.589)
Firm-years / firms	17,486 / 4,796				

*Random-effects Tobit with limits at 0 and 100, estimated in correlated random-effects form. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Firm size, business group, intra-EU sales, public funding, market scope, industry effects, year effects and the Mundlak block included but not reported. Neither contested moderation is estimable on this margin, so the falsification tests in columns 3 and 4 are adjudicated on the extensive margin and are reported here for completeness.*

On the intensive margin the High-intensity non-captive R&D purchase raises the appropriated share by 1.823 to 2.018 points depending on specification, significant at five per

cent, while collaboration is negative and non-significant throughout. Read with Table 4.8, the baseline is the shape the framework predicts on both margins.

However, neither moderation is distinguishable from zero at the intensive margin: H1 is -0.537 and H2 +0.555, with comparable nulls under the continuous dose. Section 2 anticipates this. If the disclosure hazard threatens the occurrence of market entry rather than the magnitude of return conditional on entry, the moderation belongs on the extensive margin, and its absence on the intensive margin is consistent with the micro-foundation rather than evidence against the mechanism.

The F1 interactions are estimated in a separate specification carrying both non-patent interaction terms. F2 is tested on the extensive margin only, since neither contested moderation is estimable on the intensive margin and a placebo carries no information where the treatment interaction it is tested against is itself indistinguishable from zero.

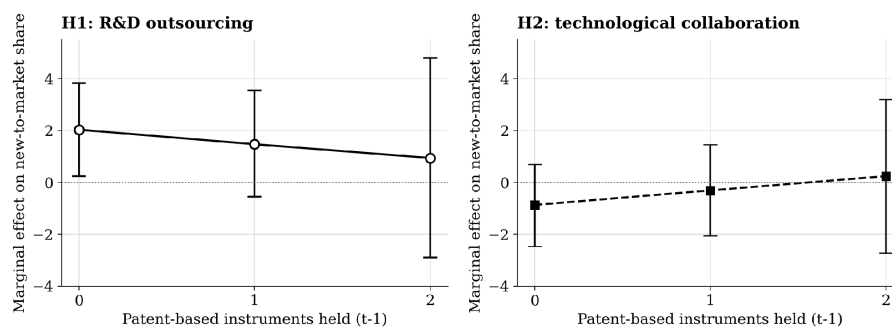


Figure 4.7. Marginal effect of each governance mode on the new-to-market share across levels of patent-based protection, with 95 per cent confidence intervals.

Figure 4.7 reports the same marginal effects on the profitability margin and should be read against Figure 4.6. Two things distinguish it: Every interval contains zero at every level of the moderator, for both channels, so nothing on this margin is estimable with the precision the extensive margin affords. And the two point-estimate profiles run in opposite directions, outsourcing falling from +2.03 to +0.94 and collaboration rising from -0.87 to +0.24, where on the extensive margin both decline together. That divergence is not evidence of anything, because neither slope is distinguishable from zero.

The bundle

Because interaction coefficients in non-linear models cannot be read directly, all cells are also estimated as mutually exclusive categories against a reference of exposed, unprotected firms.

Table 4.10. The alignment matrix estimated as exclusive categories, extensive margin

Category	Predicted probability	Row contrast	p
Reference: externally exposed, no patent-based protection	0.271	-	-
A. Non-captive purchase with patent-based protection	0.316	A - B = -0.069	0.034
B. Non-captive purchase, non-patent protection only	0.385		
C. Collaboration with patent-based protection	0.402	C - D = +0.139	0.00013
D. Collaboration, non-patent protection only	0.356		
Both modes with patent-based protection	0.345	-	-
Both modes, non-patent protection only	0.290	-	-
No external sourcing	0.218	-	-

Dependent variable: market-novel product introduction (0/1). Predicted probabilities from the exclusive-category specification, evaluated at observed values. Row contrasts are linear combinations on the index scale and are not differences of the predicted probabilities in the adjacent column.

In the market row the non-patent cell outperforms the patent cell and the contrast rejects equality, a formal refutation of the Cell A prediction that agrees in sign with the interaction estimate. In the relational row the patent cell outperforms the non-patent cell and the contrast rejects decisively, confirming the Cell C prediction and disagreeing in sign with the interaction estimate for the same channel. Section 2 anticipates the disagreement and instructs that the two be treated as correctly specified answers to different questions: the bundle conditions on exclusive categories, so its relational cell is identified off firms that collaborate, hold patents and do not purchase. What the disagreement establishes is that the relational-row (H2) verdict is population-dependent in a way the market-row (H1) verdict is not.

4.4 Appropriation as a claim over the output

Table 4.11 reports the intensive margin, decomposed into within-firm and between-firm components. The firm-mean coefficient is the difference between the between-firm and the within-firm effect rather than the between-firm effect itself, so the between-firm quantity on which H3b is stated is the sum of the two coefficients and is estimated by linear combination.

Table 4.11. Intensive margin: within-firm and between-firm components of appropriation

Construct	Within-firm coefficient	Firm-mean coefficient	Between-firm effect (sum)	Equality test, p
Patent-based protection [H3a, H3b]	+1.606** (0.707)	-0.827 (1.387)	+0.779	0.551
Non-patent formal protection	+0.980 (0.654)	-4.740*** (1.698)	-3.760	0.005
Technological collaboration	-0.656 (0.748)	-5.962*** (1.965)	-6.618	0.002
Institutional R&D purchase	-1.002 (0.682)	-3.920* (2.048)	-4.922	-

Construct	Within-firm coefficient	Firm-mean coefficient	Between-firm effect (sum)	Equality test, p
High-intensity non-captive purchase	+1.823** (0.797)	+0.608 (2.613)	+2.431	-
Absorptive capacity	+14.405*** (2.968)	+19.067*** (5.012)	+33.472	-
Firm-years / firms	17,486 / 4,796			

*Dependent variable: new-to-market sales share (%). Random-effects Tobit with limits at 0 and 100, estimated in correlated random-effects form. No governance-by-patent interaction enters this specification. The firm-mean coefficient is the difference between the between-firm and within-firm effects; the between-firm effect is their sum and is estimated by linear combination. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

H3a is supported. A contemporaneous within-firm increase in patent-based protection is associated with a 1.606-point increase in the new-to-market sales share (SE 0.707). The estimate is stable across specifications: +1.750 under the linear correlated random-effects form against a fixed-effects benchmark of +1.760.

H3b is not supported, and the failure is directional rather than a matter of precision. H3b predicted a negative between-firm association. For patent-based protection the between-firm effect is +0.779, positive and therefore opposite to the predicted sign. The formal test of equality between the stock and the flow returns $p = 0.551$, so the two components cannot be distinguished for patents.

H3 predicts a specific architecture: a positive within-firm flow, a negative between-firm stock, and the two statistically distinguishable. The non-patent placebo satisfies that architecture exactly (within +0.980, between -3.760), difference significant at $p = 0.005$, while patents do not, with both components positive and indistinguishable. The stock-flow structure Chapter 2 constructs is empirically real; it operates on the instruments that confer no claim over technical content rather than on the ones that do.

The pattern is not specific to patents. The same negative between-firm effect appears for habitual collaboration (-6.618) and habitual institutional purchase (-4.922), neither of which admits a defensive-portfolio reading. The negative between-firm sign is therefore general to habitual innovation engagement measured on a share outcome (the caution stated in advance under H3b) and is not evidence of defensive accumulation in any instrument. What this qualifies is the F1 placebo, not H3b: F1 was built for the moderation hypotheses, and there it passes. It says nothing about the output role, where the same negative sign attaches to trademarks just as it does to habitual collaboration and institutional purchase. None of these instruments defends a market position; what they share is a broad turnover base, which a share outcome mechanically penalises.

H4: margin specificity

Table 4.12 Margin specificity of the protection flow

Interaction with the market-novel margin	Pooled	Within firm and margin
Market-novel margin × patent-based protection [H4]	0.953 (0.744)	+2.897*** (0.910)
Market-novel margin × non-patent protection [F1]	-1.142 (0.770)	+0.604 (0.867)
Market-novel margin × non-captive purchase	3.150*** (1.191)	+1.820 (1.142)
Market-novel margin × collaboration	-2.485*** (0.853)	-1.097 (1.021)
Market-novel margin × institutional sourcing	-2.712*** (0.974)	-1.084 (1.099)
Patent-based protection, new-to-firm margin	-0.128 (0.383)	-1.129** (0.484)
Observations	34,972	34,972

*Dependent variables stacked: new-to-market sales share (%) and new-to-firm sales share (%). Column 1 is pooled with standard errors clustered on the firm; column 2 absorbs firm-by-margin effects, so the interaction is a difference in within-firm slopes. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$.*

H4 is supported. Table 4.12 reports the margin-specificity test on stacked outcomes. H4 is stated about a within-firm effect and therefore requires within-firm identification; the pooled specification returns 0.953 and cannot adjudicate. Absorbing firm-by-margin effects, the contrast is +2.897 ($p = 0.0014$) and the non-patent placebo remains non-significant at +0.604. Within a firm, an increase in patent-based protection is associated with -1.129 on the new-to-firm share against an implied +1.768 on the new-to-market share. One qualification applies and should be stated: the two outcomes are shares of the same turnover and are jointly bounded, so a shift in composition is partly arithmetic.

H5: absorptive capacity as complement

H5 is partially supported. The interaction between the treatment and internal research intensity is +8.309 with a standard error of 4.160 on the intensive margin, significant at five per cent, against -1.836 and non-significant for collaboration. This is the only positive, significant moderator of the treatment anywhere in the study, and it appears on the margin where the framework anticipates patents may attenuate the same relationship. The main effect of absorptive capacity is among the largest in either equation, at 14.405 on the intensive margin and 0.844 on the extensive margin.

4.5 Robustness

Robustness is organised against the specific alternative explanations the design admits rather than as an undifferentiated battery: the measurement of the treatment, the measurement of the moderator, the functional form, the estimator, and the sample.

As reported in Table A4.13 in the appendix, the moderation survives all three measures of patent-based protection. The binary indicator cannot distinguish a patent from a utility model, so the result is not an artefact of aggregating the two, and it returns the largest interactions of the three. Substituting a count of all formal instruments reproduces the pattern at reduced magnitude, which is what a composite diluting a patent effect with a non-patent one should produce.

The effect survives every specification with power to test it, including a fractional-response (Papke and Wooldridge 1996) specification resting on different distributional assumptions, and the removal of light sourcers from the reference category. Channel separation is necessary: pooling all counterparties returns 0.181 on the extensive margin, against 0.185 for non-captive commercial purchase, 0.070 for intra-group purchase and 0.027, not significant, for institutional purchase. The intensive-margin interactions are not significantly different from zero under any patent measure.

Two results conditional on measurement are reported in Table A.14 in the appendix. Substituting a human-capital measure of absorptive capacity returns -0.078 for the within-firm patent coefficient, but that substitution also restricts the sample. Varying the population and the control one at a time shows that the population is not what moves the coefficient, since the estimate on the smaller sample with the original control is +2.020. H3a therefore holds wherever internal research intensity enters the model. Because internal research intensity is plausibly on the causal path between patenting and market-novel output, this conditionality is a limitation rather than a reassurance and is taken up in Chapter 5.

As reported in Table A.15 in the appendix, the breadth and composition alternatives answer the characterisation question of 2.4. At the extensive margin the patent-specific measure is sharper than the count, so the functional distinction between instruments conferring a claim over technical content and instruments that do not is doing empirical work. At the intensive margin the two fit equally well, the log-likelihoods differing by 0.48 on equal degrees of freedom, so on that margin the distinction earns nothing. Composition is uninformative: every coefficient is non-significant on a sample restricted to IP-active firms, and the selection

correction there is an order of magnitude larger than anywhere else in the study, the signature of a strongly selected population.

Reference category

The principal specification compares heavy sourcers against everyone else, so its reference category pools firms that source lightly with firms that do not source at all. If the H1 interaction were an artefact of that pooling, restricting the comparison to the two ends of the distribution would change it. It does not. As reported in Table A.16 in the appendix, The intensive-margin estimate is likewise unchanged and remains indistinguishable from zero under either reference category. The disconfirmation of H1 at the extensive margin is therefore not a consequence of who occupies the comparison group.

4.6 Summary

The verdicts against hypotheses stated in Section 2.4 are summarised in the appendix in table A4.17. Patent-based protection shifts the composition of a firm's own revenue toward appropriable, market-novel output, and simultaneously reduces what the firm obtains from both channels of external knowledge sourcing. The two roles that Section 2.2 separates analytically are not merely distinct in these data; they carry opposite signs. That tension is more informative than a uniform confirmation would have been, since it locates appropriation's effect precisely rather than attributing to it a general facilitating role.

5. Discussion and Conclusions

5.1 Summary of the findings

This study investigates whether appropriation mechanisms condition what firms obtain from governing R&D externally, and whether the answer differs between market and relational governance. The question was posed because two established literatures give opposite answers to it, and neither has estimated the quantity on which they disagree. One tradition holds that formal protection renders external exchange more productive, since a firm that is able to defend a claim over what it discloses can afford to disclose more (Williamson 1985; Grossman and Hart 1986; Arora and Ceccagnoli 2006). Under this signaling view, policy frequently treats a strong intellectual property position as evidence of readiness to collaborate, occasionally employing patent databases as a screening mechanism to identify partners (Arora et al., 2016; Miozzo et al., 2016). Conversely, a second tradition finds that a

protective posture is incompatible with the openness on which the absorption of external knowledge depends (Arora et al., 2016; Laursen and Salter, 2014; Miozzo et al., 2016). Consequently, institutional policies designed to universally foster formal IPR usage can backfire, as forcing registered patents onto small firms can disrupt their natural innovation regimes and inhibit further knowledge sharing (Thomä and Bizer, 2013). Both traditions estimate appropriation as a determinant of the firm's sourcing decisions rather than as a condition on the returns to those decisions, so the disagreement had never been adjudicated on the quantity to which it refers. Estimating the two accounts against one another on Spanish manufacturing firms observed between 2009 and 2016 yields four findings.

Appropriation attenuates the returns to external sourcing in both governance channels.

Firms that purchase R&D at arm's length and firms that co-develop with external partners are each substantially more likely to introduce a market-novel product than firms that do neither. That advantage declines as the firm's patent position strengthens, and in the market channel it is exhausted among the most heavily protected firms. The attenuation is of similar magnitude in both outsourcing and collaboration channels; it does not arise when patent-based instruments are replaced by trademarks and copyright, nor when arm's-length purchase is replaced by purchase from within the corporate group or from public research organisations. The alignment account is therefore disconfirmed and the substitution account supported, directionally rather than through a failure of detection.

This extends the paradox of openness rather than contradicting it. Laursen and Salter (2014) establish that firms with stronger appropriability strategies engage less extensively in external search and collaboration; the evidence here concerns the firms that engage anyway, and shows that the friction they document at the point of the sourcing decision persists into the return on it. This is a stronger proposition than theirs, and one their design, in which openness is the dependent variable, is not positioned to test. It qualifies rather than refutes the secure-openness argument: the evidence is not that protection never facilitates disclosure, but that in this population the facilitating effect, if present, is outweighed. One qualification is necessary: if relational friction binds only at high legal aggressiveness, as the concavity in their estimates implies, attenuation should be confined to heavily protected firms. It does not: the sourcing advantage falls from the first instrument onward. Whatever the mechanism is, it is not confined to the aggressive tail.

Appropriation nonetheless performs its expected function, over the firm's own output rather than over its external relationships. An increase in patent-based protection is

associated with a rise in the share of turnover derived from market-novel products (radical innovation) and a decline of comparable magnitude in the share derived from products new only to the firm (incremental innovation). This is the property-rights prediction, and it survives the same falsification tests the moderation results satisfy. Together the two constitute the central finding: the same instrument, identically measured within the same firms, raises the return to internally generated innovation and lowers the return to externally obtained knowledge. Appropriation is thus neither a general enabler of open innovation nor a general obstacle to it, its effect being conditional on the provenance of the knowledge to which it applies.

The decomposition bears on an unresolved result in the literature. Hurmelinna-Laukkanen, Sainio and Jauhiainen (2008) report a strong appropriability regime raising the incremental share of turnover while returning no significant effect for radical innovation. Read through the two roles separated here, that null is not evidence that appropriation is inert at the radical margin but the expected outcome of averaging a positive output effect against a negative sourcing effect within a single coefficient.

The moderation operates on the occurrence of radical innovation, not on the revenue appropriated from it. The attenuation is estimable on the probability of introducing a market-novel product and is indistinguishable from zero on the revenue share, and the non-parametric profiles reveal no divergence between protected and unprotected firms in any populated region of the sourcing distribution. This is what the disclosure argument anticipates: leakage enables a rival to match the firm's timeline or reach the market first, but does not, in the first instance, weaken the margin on a product already commercialised under the firm's name. The result is therefore a confirmation of the micro-foundation rather than bounding it, though it entails that the thesis speaks to the occurrence of radical innovation with more authority than to its profitability.

The complement to external sourcing is internal research capability rather than legal protection. Where arm's-length purchase is combined with high internal research intensity the appropriated share rises, while the equivalent combination with collaboration returns no significant effect. Internal research capability is accordingly the only condition examined that raises rather than lowers the return to external sourcing, which recovers the original absorptive-capacity argument (Cohen and Levinthal 1990) in a setting where the alternative candidate for that role fails. The asymmetry between channels is not explained by the framework and is reported rather than claimed as a finding.

One prediction failed informatively. The thesis anticipated that firms with habitually large patent portfolios would exhibit a lower market-novel revenue share, following the defensive-patenting literature. The between-firm association is positive, and the accumulated stock cannot be distinguished from the current act of patenting. Negative between-firm associations do appear, for habitual collaboration and habitual institutional sourcing, but they attach to activities carrying no defensive interpretation. The pattern is therefore a property of measuring an outcome as a share of turnover among habitually innovative firms rather than evidence of defensive accumulation. The stock and flow distinction might be analytically sound; it simply does not operate on patents in these data.

5.2 Contributions

Appropriation is repositioned from a determinant of governance to a moderator of performance. In the literature reviewed, appropriation enters the right-hand side of an equation whose dependent variable is a sourcing decision or the breadth of a firm's openness; whether firms sourcing under a strong regime perform better is estimated in neither. This thesis moves the construct to the interaction term and asks whether protection conditions the productivity of external exchange rather than the propensity to engage in it. That is a different question, and it is the one on which the two accounts genuinely diverge, and posed directly it returns an answer neither literature had established.

The two roles of appropriation are separated and shown to carry opposite signs. Existing work treats appropriation as a single construct with a single directional value. Distinguishing the claim over the firm's own output from the condition on external exchange, and estimating both for the same instrument in the same firms, demonstrates that the construct is not directionally coherent. A composite index of appropriability, of the kind widely used, averages the two and will understate whichever is the larger. This is a measurement implication as much as a theoretical one.

A two-channel design arbitrates between accounts a single interaction cannot separate. The two accounts differ not only in predicted sign but in the cross-channel pattern implied: alignment reaches the market and relational channels through distinct mechanisms and permits the magnitudes to differ, whereas substitution operates on the protective posture itself and implies a common attenuation. Estimating both channels within one specification renders their difference an estimable quantity, converting a question about signs into a question about

patterns. The design is portable to any setting in which rival accounts imply the same sign through different mechanisms.

The falsification architecture is instrument-specific and channel-specific rather than generic. Two placebos are derived from the theory rather than chosen for convenience: instruments conferring no claim over technical content should moderate nothing, and channels in which the disclosure hazard is contained by ownership or publication norms should carry no patent moderation. Each is assessed by a test of equality against the corresponding treatment interaction rather than against zero, since a difference in significance is not a significant difference. This raises the evidentiary standard above what a robustness battery achieves: the tests are constructed so that they could have refuted the reading rather than merely failing to disturb it.

The two-margin decomposition locates the effect rather than averaging over it. Estimating the occurrence of market-novel innovation and the revenue appropriated from it as distinct outcomes establishes that the moderation is a phenomenon of market entry and not of profitability. A design that collapsed the margins into a single share outcome would have recovered a weak average of an effect that is strong on one margin and absent on the other, and would have supported a materially weaker conclusion.

5.3 Policy implications

Spanish and European innovation policy supported patenting and external collaboration through largely separate instruments across the observation window: the Spanish patent office ran annual fee-subsidy programmes carrying no collaboration condition (OEPM 2014, 2015), while collaborative R&D was funded through consortium-based schemes with no reference to the applicant's protective position (CDTI), so whether they interact, and with what sign, goes unexamined (Flanagan, Uyarra and Laranja 2011). The evidence here does not support that premise. Each implication below is stated for the population studied and rests on a conditional association rather than an identified causal effect.

Table 5.1. Policy implications and the finding on which each rests

Recommendation	Content	Supporting finding
Do not bundle IP support with collaboration requirements	Programmes making collaboration funding conditional on a demonstrated intellectual-property position combine two instruments that work against one another on the entry margin	Negative moderation in both channels, robust across three measures of patent-based protection
Support absorptive capacity alongside external sourcing	Subsidies for acquiring external R&D should be paired with support for the internal research capability that converts it into commercialised product, not treated as substitutes for it	Internal research intensity is the only positive moderator of external sourcing estimated
Set a minimum scale on acquisition subsidies	Small subsidised purchases fall below the commitment threshold above which external sourcing raises the appropriated share, and generate no measurable return	Threshold recovered non-parametrically at the median intensity among purchasing firms
Treat public research partnerships as the low-friction channel	Sourcing from universities and public research organisations exhibits no attenuation under patent protection, so technology-transfer policy is unaffected by the tension identified	Institutional placebo passes; no patent moderation in that channel

Each recommendation is bounded by the population studied and by the associational standing of the underlying estimates.

The first implication is the least intuitive and the most consequential. Policy treats an intellectual-property position as a screening criterion in its own right, for example the Horizon 2020 SME Instrument requiring single-firm applicants to evidence IP ownership and freedom to operate (European Commission 2018). On these estimates, the firms selected by that criterion are those deriving the least benefit from collaborating. The criterion is not mistaken about firm quality, since heavily patenting firms innovate more in absolute terms, but it selects on the wrong margin if the objective is to raise the productivity of the partnership itself. A caution is attached to all four: these results identify a tension; they are not an argument for weakening intellectual-property protection. The output finding runs in the opposite direction, and a policy discouraging patenting would forfeit it. What the evidence argues against is the assumption that a single instrument serves both purposes.

5.4 Limitations

Identification rests on the structure of the panel and the timing of the variables, not on an exogenous shock or an external instrument. Every estimate is therefore a conditional

association consistent with the theorised mechanism rather than an identified causal effect, since a firm's protective posture and its sourcing behaviour may both respond to an unobserved technological position. Relatedly, the proportion of firms changing treatment or collaboration status within the window is limited, so the moderation results draw more heavily on differences between firms than on transitions within them; the output and margin-specificity results, which rest on genuine transitions, carry correspondingly greater weight.

Particular findings are bound by measurement limitations. Both revenue outcomes are percentages of the same turnover and are jointly bounded, so a shift in composition between them is in part arithmetic; this bears most directly on the margin-specificity result, for which a check on unscaled sales levels is required before the finding can be asserted at full strength. The within-firm patent effect obtains only where internal research intensity enters the model, and since that variable plausibly lies on the causal path between patenting and market-novel output, conditioning on it may bias the estimate rather than purge it. Patent-based protection is finally measured as a count of instrument use rather than as portfolio strength or enforcement intensity, which is the dimension along which the concavity argument operates, so the moderator cannot establish where on that curve a firm is located.

The population comprises Spanish manufacturing firms under a single institutional regime. The findings should not be extended to services, to economies with materially different enforcement capacity, or to periods of substantial reform in intellectual-property law.

5.5 Directions for further research

Four extensions follow from these limitations. The first is the separation of domestic from foreign arm's-length purchase, since cross-border enforcement asymmetry is a plausible source of variation in the severity of the disclosure hazard and the closest precedent establishes its effect specifically for acquisition from abroad. The second is the measurement of protection intensity rather than instrument count, permitting a direct test of the concavity implied by the paradox of openness; the monotone attenuation observed here is consistent with a friction operating from the first instrument, but cannot be distinguished from a concave relationship measured too roughly. The third is the two-dimensional treatment of absorptive capacity proposed by Arbussa and Coenders (2007), under which scanning capability assists all firms while integration capability assists only those that protect; if that

holds, it would supply a mechanism for the channel asymmetry in the complementarity result, which this thesis reports without explaining.

A fourth question was pursued and could not be resolved on this data. This thesis measures appropriation exclusively through formal instruments, yet the survey evidence has consistently found informal mechanisms (secrecy, lead-time advantage and design complexity) to be rated more effective than patents across most manufacturing sectors (Levin et al. 1987; Cohen, Nelson and Walsh 2000; Arundel 2001; Hall et al. 2014), and firms are known to substitute towards them where formal protection is weak or costly (Thomä and Bizer 2013; Miric, Boudreau and Jeppesen 2019). The omission is consequential, and it also forgoes a sharper test than the formal instruments permit. The two families differ in visibility: a patent portfolio is observable to a prospective partner and communicates a posture, whereas secrecy and lead time protect without announcing themselves. If the attenuation documented here operates through the posture that formal protection signals, as the common cross-channel magnitude suggests, informal mechanisms should not attenuate the returns to external sourcing; if it reflects a general defensive orientation of which formal instruments are one expression among several, they should. The survey does not record the strategic protection methods over the observation window, so the prediction cannot be tested directly. Two proxies were constructed and estimated (one based on the tacit character of the firm's knowledge base, the other a residual capturing protective activity not accounted for by the formal instruments) and neither returned interpretable estimates. Since a null obtained from an indirect proxy is uninformative about the underlying mechanism rather than evidence against it, the question is left open. Resolving it requires an appropriation measure constructed for the purpose rather than recovered from a general innovation survey.

5.6 Conclusion

This thesis asked whether appropriation assists or impedes the firm that sources knowledge externally. The answer is that it does both, and that the two effects operate on different parts of the innovation process: protection secures what a firm generates internally and diminishes what it draws from others. Because the two have rarely been measured together, the literature has divided into traditions each describing one half of a single instrument. Separating the roles resolves that division without adjudicating it in favour of either, and locates the appropriation effect with some precision, on the occurrence of market-novel innovation rather than on the revenue earned from it, and on the market and relational channels alike. For the

firm, the implication is that a protective posture and an open sourcing strategy are not combined without cost. For policy, it is that the two cannot be promoted as though they were one. Finally, firms guard the gates of innovation with the same instruments they use to profit from what passes through them, and these two functions pull against each other. What makes external knowledge pay is not the strength of the gate but the capability that waits behind it.

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Any remaining errors are my own.

Declaration on the use of artificial intelligence

During the preparation of this thesis, the author used Anthropic's Claude, a large-language-model assistant, to support the following tasks, each carried out under the author's direction and subject to the author's review: summarising the reviewed literature through an interactive dashboard for quick access to references and complex syntheses; improving the clarity and concision of successive drafts; helping to articulate the theoretical framework and enhancing the presentation of the results; formatting tables; and supporting the successive iterations of the Stata do-files, which were created and adjusted at multiple iterations under the author's direction. The research design, the analytical choices, the interpretation of the results, and the hypotheses and conclusions are the author's own. All AI-assisted text, code and output was reviewed, verified against the underlying sources and data, and edited as required, and the author takes full responsibility for the content of this thesis.

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Data source

Instituto Nacional de Estadística (INE), Fundación Española para la Ciencia y la Tecnología (FECYT) and Fundación COTEC. Panel de Innovación Tecnológica (PITEC), waves 2003-2016. Madrid: INE

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A. Chapter 2

Table A2.4. Hypothesis summary, with the disconfirming pattern for each

	Claim	Predicted pattern	Disconfirming pattern
Base.	Both governance modes raise market-novel innovation on both margins	Positive participation effect subject to a hollowing-out ceiling; intensive-margin threshold	Null or negative participation effect; no threshold structure on the intensive margin
H1	Patents positively moderate outsourcing	Marginal effect of outsourcing rising in patent protection	Flat or declining marginal effect; a decline confirms substitution
H2	Patents positively moderate collaboration	Marginal effect of collaboration rising in patent protection	Flat or declining marginal effect; a decline in both channels confirms substitution
H3a	Patent flow raises the appropriated share within firms	Positive within-firm coefficient with firm effects present	Null or negative within-firm coefficient
H3b	Patent stock is negatively associated with the appropriated share between firms	Negative between-firm effect, read as a weak signature	Positive between-firm effect
H4	The appropriation effects are specific to the radical margin	Attenuation on the new-to-firm share	Equal magnitude on both margins
H5	Absorptive capacity complements external sourcing	Marginal effect of sourcing rising in internal research intensity	Flat or declining marginal effect across capability levels
F1	Non-patent formal instruments moderate neither channel	Null interactions, statistically distinguishable from the patent interactions	Interactions indistinguishable from those on patents
F2	Captive and institutional sourcing carry no patent moderation	Null interactions on the margin where the effect is precisely estimated	Moderation of comparable magnitude in the captive channel

F1 and F2 are falsification tests rather than hypotheses.

B. Chapter 3

Table A3.1. Estimation samples

Sample	Firm-years	Firms	Definition
Extensive	69,034	10,946	Complete data on all model variables
Intensive	17,486	4,796	Extensive sample, conditional on a market-novel launch
Comparison	34,972	4,796	Intensive sample with both share outcomes stacked
Human capital	14,302	4,026	Intensive sample with data on research-staff qualifications
Composition	6,909	2,417	Intensive sample, firms holding at least one formal instrument

The final two samples are used only where the relevant variable is required, and estimates obtained on them are not generalised beyond the populations they describe.

Table A3.2. Operationalisation of every construct

Construct	Variable	Operationalisation	Role
Radical innovation, extensive	novedad	1 if the firm introduced a product new to the market	Outcome, occurrence margin
Radical innovation, intensive	newmer	Percentage of turnover from products new to the market, [0,100]	Outcome, appropriation margin
Incremental innovation	newemp	Percentage of turnover from products new to the firm only	Comparison outcome (H4)
Non-captive outsourcing	noncap_hi	1 if non-captive commercial R&D purchase exceeds the median intensity among positive sourcers	Treatment
Sourcing intensity	dose	Non-captive commercial R&D purchase as a percentage of total innovation expenditure	Continuous treatment
Captive sourcing	captive	1 if the firm purchases R&D within its corporate group	Placebo channel (F2)
Institutional sourcing	inst	1 if the firm purchases R&D from universities or public research organisations	Control channel (F2)
Technological collaboration	coop	1 if the firm co-operated on innovation with an external partner	Second governance mode
Patent-based protection	patent_prot	Count in [0,2] of patent and utility-model use	Moderator (H1, H2); output claim (H3)
Non-patent formal protection	other_formal	Count in [0,2] of trademark and copyright use	Placebo moderator (F1)
Breadth of protection	formal_breadth	Count in [0,4] of distinct formal instruments	Robustness
Composition of protection	patent_share	Patent share of the instrument set, for firms holding at least one	Robustness

Construct	Variable	Operationalisation	Role
Patent flow	within-firm deviation	Deviation of patent_prot from the firm mean	H3a
Patent stock	between-firm effect	Sum of the within-firm coefficient and the firm-mean coefficient	H3b
Absorptive capacity	ln_abscap	ln(1 + internal R&D expenditure over turnover)	Control; moderator (H5)
Absorptive capacity, human capital	abscap2_pct	Share of R&D personnel with doctoral or graduate qualifications	Robustness, sub-sample
Firm size	ln_size	Log of the number of employees	Control
Business group	grupo	1 if the firm belongs to a business group	Control
Intra-EU sales	export_pp	Sales within the European Union as a percentage of turnover	Control
Public funding	pubfund_any	1 if the firm received any public innovation funding	Control
Market scope	mtonac, mtonue, mtonp	Indicators for the national, EU and other markets served	Controls
Industry	NACE 2-digit	Industry classification	Controls
Year	wave	Survey wave	Controls
Exclusion restriction	weak demand	1 if the firm cited insufficient market demand as a reason for not innovating	Selection equation

Table A3.3. Coefficient glossary

Term	What it identifies	Reads on
α_1	Main Effect of heavy non-captive purchase on the probability of a market-novel launch, for a firm holding no patent-based instrument	Baseline
α_2	Main Effect technological collaboration on the probability of a market-novel launch, for a firm holding no patent-based instrument	Baseline
α_3	Effect of protection itself on the probability of a - launch, for a firm doing neither	-
α_4	Change in the purchase effect per additional patent-based instrument held	H1
α_5	Change in the collaboration effect per additional instrument held	H2
β_1, β_2	Effect of each governance mode on the new-to-market turnover share, at zero protection	Baseline
β_3	Within-firm effect of protection on the share: the flow	H3a
θ_p^β	Between-firm effect minus within-firm effect; a formal test of whether stock and flow differ	H3 (Hausman)

Term	What it identifies	Reads on
$\beta_3 + \theta_p^\beta$	Between-firm effect of protection: the stock, by linear combination	H3b
β_4, β_5	The same two moderations on the profitability margin	H1, H2
β_6	Change in the purchase effect per unit of internal research intensity	H5

C. Chapter 4

Table A4.2. Correlation matrix and variance inflation factors, extensive-margin sample

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	VIF
(1) Market-novel launch	1													
(2) New-to-market share	0.631	1												
(3) New-to-firm share	0.062	0.037	1											
(4) Non-captive purchase	0.101	0.076	0.057	1										1.04
(5) Intra-group purchase	0.104	0.047	0.050	0.078	1									1.09
(6) Institutional purchase	0.205	0.107	0.068	0.133	0.142	1								1.30
(7) Collaboration	0.313	0.184	0.123	0.147	0.176	0.383	1							1.42
(8) Patent-based protection	0.252	0.161	0.067	0.089	0.087	0.203	0.199	1						1.25
(9) Non-patent protection	0.199	0.113	0.059	0.086	0.046	0.141	0.157	0.393	1					1.20
(10) Absorptive capacity	0.196	0.220	0.077	0.062	0.017	0.179	0.244	0.138	0.094	1				1.21
(11) Log employees	0.033	-.029	0.012	0.019	0.139	0.065	0.093	0.065	0.069	-.202	1			1.32
(12) Business group	0.051	-.004	0.025	0.012	0.230	0.064	0.134	0.045	0.039	-.096	0.451	1		1.31
(13) Public funding	0.310	0.202	0.124	0.138	0.127	0.415	0.466	0.230	0.164	0.327	0.028	0.047	1	1.50
Mean VIF														1.26

Listwise correlations on the extensive-margin sample of 69,034 firm-years, so the matrix describes the estimation sample exactly. All coefficients are significant at the 0.1 per cent level except log employees with the new-to-firm share and with intra-group purchase (both $p < 0.01$) and business group with the new-to-market share, which is not significant. The strongest association among the regressors is 0.466, between collaboration and public funding. Variance inflation factors are computed on the auxiliary specification of Table A4.2 and run from 1.04 to 1.50 with a mean of 1.26; no regressor is materially collinear with the remainder. Intra-EU sales, the market-scope indicators and the industry and year effects do not enter the diagnostic.

Table A4.3. Auxiliary regression underlying the collinearity diagnostic

Regressor	Coefficient	SE
High-intensity non-captive R&D purchase	0.049***	(0.006)
Technological collaboration	0.164***	(0.004)
Intra-group R&D purchase	0.071***	(0.008)
Institutional R&D purchase	0.025***	(0.005)
Patent-based protection (0–2)	0.136***	(0.004)
Non-patent formal protection (0–2)	0.093***	(0.004)
Absorptive capacity	0.255***	(0.013)
Log employees	0.001	(0.001)
Business group member	0.008**	(0.003)
Public innovation funding	0.140***	(0.004)
Constant	0.104***	(0.004)
Observations	69,034	
R-squared	0.173	

Linear specification on the extensive-margin sample, estimated to obtain the variance inflation factors reported in Table A4.2. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4.4. The correlated random-effects specification against fixed and random effects

<i>Dependent variable: new-to-market sales share (%).</i>	FE lagged	RE lagged	FE contemp.	RE contemp.	Mundlak RE
High-intensity non-captive purchase (t-1)	1.324* (0.752)	1.721** (0.683)			1.719** (0.769)
Patent-based protection (t-1)	1.760*** (0.503)	1.390*** (0.440)			1.614*** (0.603)
Non-patent formal protection (t-1)	1.069** (0.535)	0.323 (0.481)			0.977* (0.578)
Technological collaboration (t-1)	-0.140 (0.579)	-1.042** (0.507)			-0.435 (0.664)
Absorptive capacity (t-1)	14.381*** (2.578)	21.747*** (1.850)			13.315*** (3.577)
Intra-group R&D purchase (t-1)	-1.158 (1.029)	-0.835 (0.919)			-1.090 (1.026)
Institutional R&D purchase (t-1)	-0.780 (0.652)	-1.573*** (0.581)			-0.870 (0.724)
High-intensity non-captive purchase (t)			0.323 (0.819)	0.682 (0.742)	
Patent-based protection (t)			2.080*** (0.515)	1.512*** (0.451)	
Firm-years	17,486	17,486	17,464	17,464	17,486
Firms	4,796	4,796	4,792	4,792	4,796

<i>Dependent variable: new-to-market sales share (%).</i>	FE lagged	RE lagged	FE contemp.	RE contemp.	Mundlak RE
Joint test on firm means, p					1.65e-10

*Year effects in every column; firm means in column 5 only. The joint test on the firm means rejects the plain random-effects specification, so the firm effect is correlated with the regressors. The test is an auxiliary-regression test on the Mundlak terms and is reported for column 5 alone. Columns 3 and 4 are estimated on a marginally smaller sample and are not directly comparable in magnitude with columns 1, 2 and 5. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table A4.5. Fixed effects against the two Mundlak forms

	FE benchmark	Mundlak, construction window	Mundlak, estimation sample
High-intensity non-captive purchase (t-1)	1.3242* (0.7518)	1.7187** (0.7194)	1.2985* (0.7485)
Patent-based protection (t-1)	1.7598*** (0.5030)	1.6143*** (0.4819)	1.7501*** (0.5009)
Non-patent formal protection (t-1)	1.0691** (0.5353)	0.9771* (0.5136)	1.0579** (0.5332)
Technological collaboration (t-1)	-0.1399 (0.5786)	-0.4348 (0.5495)	-0.1238 (0.5761)
Absorptive capacity (t-1)	14.3811*** (2.5775)	13.3146*** (2.4375)	14.3746*** (2.5675)
Intra-group R&D purchase (t-1)	-1.1584 (1.0294)	-1.0899 (0.9940)	-1.1586 (1.0255)
Public innovation funding (t-1)	-1.8140*** (0.5655)	-1.4555*** (0.5394)	-1.8279*** (0.5632)
Largest absolute gap against FE	-	1.0665	0.0360

*Linear panel estimation of the new-to-market share on an identical sample of 17,486 firm-years and 4,796 firms, with year effects. The three columns differ only in the treatment of the firm effect and, between columns 2 and 3, in the window over which the firm means are computed. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

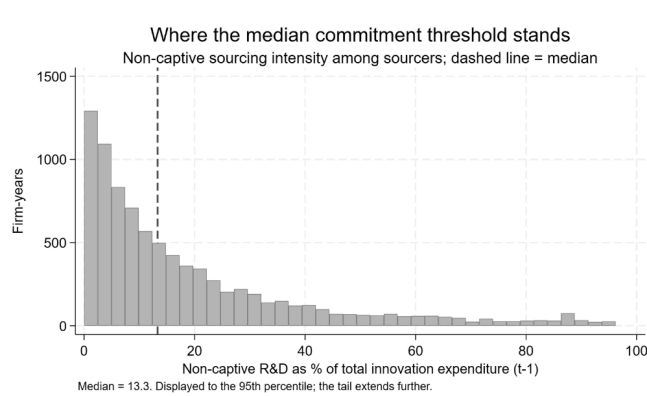


Figure A4.1. Distribution of non-captive sourcing intensity among firms that source, with the median marked. Displayed to the 95th percentile; the tail extends further.

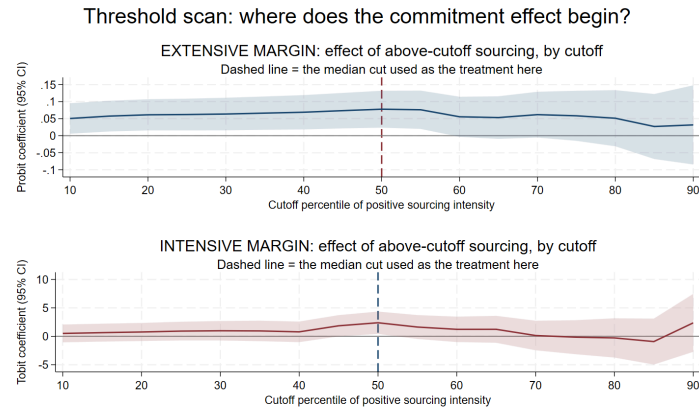


Figure A4.2. The treatment coefficient estimated at candidate cutoff percentiles, both margins, with 95 per cent confidence intervals. The dashed line marks the median cut used as the treatment.

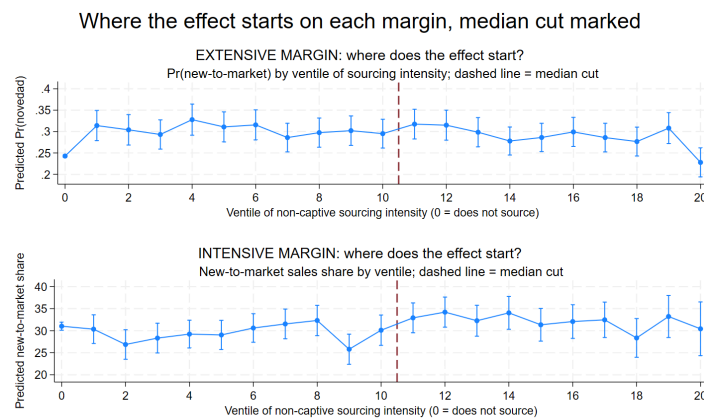


Figure A4.3. Predicted outcome by ventile of sourcing intensity, both margins, with the median cut. Ventile 0 is firms that do not source.

Table A4.6. Sourcing-intensity profile on the reference control set

Group	Extensive margin (probit)	Intensive margin (Tobit)
Decile 1	0.246***	-2.327*
Decile 2	0.250***	-2.157*
Decile 3	0.261***	-1.125
Decile 4	0.184***	+0.932
Decile 5	0.209***	-3.079**
Decile 6	0.271***	+2.565**
Decile 7	0.171***	+2.077
Decile 8	0.188***	+0.620
Decile 9	0.147***	-0.432
Decile 10	0.102**	+1.333
Ventile 9	-	-5.226***
Ventile 12	-	+3.196*
Ventile 20 (top 5 per cent)	-0.060 (0.072), ns	-

Reference category: firms that do not source. Estimated on the reference control set, carrying the appropriation, absorptive-capacity and Mundlak blocks, so the magnitudes are comparable with the estimated models. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4.7. Selection-stage probit and the exclusion restriction

Quantity	Full candidate set	Single retained instrument
Insufficient market demand as a reason not to innovate	-0.163***	-0.160***
Previous innovations sufficed	+0.004 (0.0089), ns	not retained
Exclusion test in the outcome equation, p	0.447	0.320
Within-firm patent flow under this correction	+1.606 (p = 0.023)	+1.613 (p = 0.022)
Within-firm patent flow with no correction	+1.742 (p = 0.0009)	+1.742 (p = 0.0009)

Dependent variable: market-novel product introduction (0/1). Estimated on the extensive-margin sample of 69,034 firm-years, carrying the model covariates and the firm-mean block. *** $p < 0.01$.

Table A4.13. Robustness analysis

Specification	H1 interaction (extensive)	H2 interaction (extensive)	Purchase effect (intensive)
Principal specification	-0.208*** (0.039)	-0.150*** (0.025)	2.018** (0.914)
R1. Patent applications, log count	-0.250***	-0.106***	1.684*
R2. Any patent-based instrument (binary)	-0.304***	-0.178***	1.536*
R3. Selection correction applied	-	-	2.018**
R4. Breadth of formal protection substituted	-0.132***	-0.076***	2.065**
R5. Fractional-response specification	-	-	0.085***
R6. Without correlated random effects	-	-	2.079**
R7. Non-sourcers as sole reference category	-0.216***	-	-0.353

Dependent variables: market-novel product introduction (0/1) for the extensive-margin columns; new-to-market sales share (%) for the intensive-margin column. Standard errors in parentheses where reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4.14. The within-firm patent flow, sample and control varied one at a time

Specification	C1 ln_abscap Full sample	C2 ln_abscap HC sample	C3 abscap2 HC sample
Patent-based protection (0-2) (t-1) [H3a]	1.606**	2.020**	-0.078
Firm mean: patent-based protection [H3b]	-0.827	-0.472	0.046
Observations	17,486	14,302	14,302

Table A4.15. Breadth and composition as alternatives to the patent measure

Quantity	Patent measure	Breadth count	Composition share
H1 interaction, extensive margin	-0.213***	-0.132***	not estimated
H2 interaction, extensive margin	-0.140***	-0.076***	not estimated
Treatment, intensive margin	2.018**	2.065**	1.495, ns
Appropriation main effect, intensive margin	1.305, ns	1.267*	-0.500, ns
Inverse Mills ratio, intensive margin	-0.981	-	-7.943***
Log-likelihood, intensive margin	-77,749.64	-77,750.12	-
Observations, intensive margin	17,486	17,486	6,909

*Breadth is a count of distinct formal instruments in [0,4]; composition is the patent share of the instrument set, defined only for firms holding at least one instrument. Neither may enter a model alongside its own components. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table A4.16. The H1 interaction under two reference categories

Specification	Reference category	Extensive	Intensive
Principal	Light sourcers and non-sourcers pooled	-0.209*** (0.039)	-0.537 (1.123)
Clean contrast	Non-sourcers only; light sourcers dropped	-0.216*** (0.040)	-0.353 (1.272)
Clean contrast, reconciled threshold	Non-sourcers only; cut on the estimation-sample median	-0.211*** (0.039)	-0.356 (1.231)

*Extensive-margin p -values are 8.2×10^{-8} and 7.8×10^{-8} for the two clean-contrast rows. Dropping light sourcers moves the interaction by 0.007 and rebuilding the threshold moves it back by 0.005, in both cases well inside the standard error. *** $p < 0.01$.*

Table A4.17. Verdicts against the hypothesis set of Table 2.4

Claim	Verdict	Evidence
Baseline, extensive margin	Supported	Both modes positive at 1%, non-monotonic in intensity (Table 4.8)
Baseline, intensive margin	Partially supported	Threshold holds for purchase (+2.018**); collaboration not significant (Table 4.9)
H1: patents positively moderate purchase	Not supported; substitution instead	-0.208*** (Table 4.8)
H2: patents positively moderate collaboration	Not supported; substitution instead	-0.150*** (Table 4.8)
H3a: patent flow raises the appropriated share	Supported	+1.606** within firms (Table 4.11)
H3b: patent stock negative between firms	Not supported	Between-firm effect +0.779; stock and flow indistinguishable (Table 4.11)
H4: margin specificity	Supported	+2.897*** within firm and margin (Table 4.12)

Claim	Verdict	Evidence
H5: absorptive capacity complements sourcing	Partially supported	+8.309** for purchase; -1.836 ns for collaboration (Table 4.9)
F1: non-patent placebo	Passed	Not significant in either channel on either margin (Tables 4.8, 4.9)
F2: captive and institutional placebo	Passed, qualified	Null on the extensive margin; intra-group interaction -2.380* on the intensive margin (Tables 4.8, 4.9)

Summary from Tables 4.8 to 4.13.